Tab 1

A Digital Signature Schemes Without Using One-way Hash and Message Redundancy and Its Application on Key Agreement

1. Comprehensive Taxonomy Integration

The paper defines a specific sub-category within your research: **Hash-less Integrity Schemes**.

- **Methodology (Cryptographic Assumptions):**
  - **Hash-Based (Standard):** Uses one-way functions (SHA-256) and message redundancy.
  - **Hash-less (Non-Standard):** Schemes like Chang-Chang (2004) that rely solely on public-key algebraic properties.
- **Application Domain:**
  - **Key Agreement Protocols:** Used to establish secure communication channels without heavy computational overhead (suitable for mobile/resource-constrained environments).
  - **Identity Authentication:** Verifying users without storing large hash tables.

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2. Comparative Technical Analysis

Based on the paper's findings, we can compare the **Chang-Chang Scheme** (the target of the attack) with the **Improved Scheme** proposed by Zhang et al.

Metric	**Chang-Chang Scheme (2004)**	**Zhang et al. Improved Scheme (2007)**
Integrity Mechanism	Pure Algebraic (No Hash)	Improved Algebraic + Discrete Log
Attack Vulnerability	**Existential Forgery** (Chien's Attack & Zhang's Attack)	Secure against Weak Chosen Message Attack
Complexity	$O(n)$	$O(n \log n)$ (due to exponentiation)
Key Size/Efficiency	High (No redundancy/padding)	High (Maintains no-hash advantage)
Security Assumption	Discrete Logarithm	Decisional Diffie-Hellman (DDH)

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3. Critical Gap Identification

Based on this paper, the "Critical Gap" in current literature is:

- **The Vulnerability of "Efficiency over Security" (The Redundancy Gap):** The paper highlights that while removing one-way hash functions and message redundancy makes protocols more efficient for mobile systems, it creates a mathematical "slack" that attackers can exploit to forge signatures.
- **The Oracle Problem in Signature Verification:** Most schemes assume a "Random Oracle" for security proofs. This paper identifies a gap where schemes failing to account for specific forgery attacks (like the one Zhang found) remain insecure even if they appear mathematically sound on the surface.

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4. Evolution of Solutions: Narrative Flow

This paper fits perfectly into the **Phase 2 and Phase 3 transition** of your narrative:

- **Early Phase (2004):** Designers (Chang & Chang) attempted to simplify digital signatures by removing "unnecessary" components like one-way hashes to save on communication costs.
- **Attack Phase (2006-2007):** Researchers (Chien, then Zhang et al.) proved that these simplified metadata/signature structures are inherently vulnerable to **Existential Forgery**, where an attacker can craft a valid-looking signature for a message without knowing the private key.
- **Modern Phase (Current):** The field has moved toward "Hybrid Security," where researchers like Zhang propose protocols that remain efficient (no hash) but add algebraic complexity to ensure that the "Forensic Integrity" of the signature cannot be compromised by simple

2.)forge-and-substitution attacks. LSB Steganography Detection Algorithm

1. Comprehensive Taxonomy Integration

This paper contributes to the **Detection & Steganalysis** branch of your taxonomy. It differentiates between specific "Cover-Object" types and the statistical methods used to uncover hidden data.

- **Methodology (Detection Type):**
  - **Blind Detection (GEFR):** The proposed method (Gradient Energy-Flipping Rate) which does not require the original image to detect forgery.
  - **Signature-Based/Fingerprint Detection:** Methods like JPEG compatibility checks (Fridrich et al.) that look for specific artifacts left by compression.
- **Application Domain:**
  - **High-Resolution Raw Scans:** Images where unique color counts exceed 50%, making older methods like RQP (Raw Quick Pairs) unreliable.
  - **Digital Camera Images:** Uncompressed formats where metadata is most vulnerable to sophisticated steganographic forgery.

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2. Comparative Technical Analysis

The paper provides a technical critique of existing SOTA (State-of-the-Art) tools circa 2003, comparing them against the newly proposed **GEFR method**.

Feature / Method	RQP Method (Fridrich)	RS Method (Dual-Detection)	GEFR Method (Proposed)
Detection Basis	Close pairs of colors	Probability Statistics	**Gradient Energy Variation**
Grayscale Support	No	Yes	**Yes**
High-Res Accuracy	Unreliable (>50% colors)	Moderate	**High (Effective at 0.01 bpp)**
Blind vs. Non-Blind	Blind	Blind	**Blind**
Computational Cost	Low	High	**Low (Spatial Domain)**

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3. Critical Gap Identification

The authors identify a specific "Forensic Gap" that was present in literature at the time:

- **The "Natural Image" Complexity Problem:** Most early steganography detection relied on images with a low number of colors (palette images). However, "natural" images (like those from digital cameras) have a near-random LSB distribution.
- **The Original Image Dependency:** The authors cite that previous gradient-based methods (Guo-Shiang Lin) required the *original* image to detect a change. In a real-world forensic investigation, the "original" is almost never available, creating a gap for **Blind Forensic Tools**.

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4. Evolution of Solutions: Narrative Flow

This paper represents the transition from **Simple Statistical Checks** to **Advanced Spatial-Domain Analysis**:

- **The Palette Era (Pre-2000):** Steganalysis focused on finding "clusters" of colors in low-quality images.
- **The Statistical Era (2001-2003):** Methods like RS and RQP began using probability to predict the LSB plane based on the other 7 bits.
- **The Gradient Energy Era (Li Zhi, 2003):** This paper introduces the concept that steganography alters the "smoothness" or **Gradient Energy** between adjoining pixels. This marked a shift toward analyzing the *physical characteristics* of the digital artifact rather than

just the mathematical distribution of bits. This is a precursor to modern "Error Level Analysis" (ELA) often used today to detect metadata and image forgery.

3.)A PUF-Based Data-Device Hash for Tampered Image Detection and Source Camera Identification

1. Comprehensive Taxonomy Integration

This paper establishes a new high-level category in your taxonomy: **Hardware-Assisted Forensics**.

- **Methodology (Integrity Type):**
  - **Perceptual Hashing:** Focuses on the "appearance" of the image. It allows for benign changes (like resizing) but flags malicious changes (like face-swapping).
  - **PUF-Defined Random Space:** Uses the unique "fingerprint" of the camera's hardware to encrypt the hash. Even if an attacker knows the algorithm, they cannot recreate the hash without the physical camera.
- **Application Domain:**
  - **Anti-Deepfake Forensics:** Specifically mentions combating "Deepfakes" and AI-driven facial swaps.
  - **Non-Repudiable Attestation:** Providing evidence in a court of law that a specific camera *must* have taken a specific photo at a specific time.

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2. Comparative Technical Analysis

You can now update your comparison table with these high-performance metrics. This paper outperforms almost all previous "Software-Only" methods.

Metric	**Software Steganalysis (Li Zhi, 2003)**	**PUF-Based Hash (Zheng et al., 2020)**
Tamper Detection Rate	~90%	**95.42%**
Source Identification	N/A (Cannot identify camera)	**100% Accuracy**
Robustness (Rotation)	Low/Moderate	**96.25% (up to 360°)**

Robustness (Scaling)	Low/Moderate	**90.42% (at 0.5x scale)**
Security Mechanism	Gradient Energy Analysis	**Bernoulli Random Space + Hardware PUF**

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3. Critical Gap Identification

The authors identify a massive "Forensic Gap" that exists in the other 39 papers of your survey:

- **The Identification-Localization Conflict:** Previous literature could either identify the *camera* (source forensics) OR locate the *tamper* (content forensics), but almost never both simultaneously.
- **The Secret Key Vulnerability:** Most perceptual hashes rely on a shared secret key. If the key is leaked, all previously "secured" images can be forged. By using a PUF, the "key" is the physical hardware itself—it cannot be "leaked" via software.

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4. Evolution of Solutions: Narrative Flow

This paper represents the **Final Evolution (Phase 4)** of your research narrative:

- **Phase 1:** Simple EXIF editing (Manual forgery).
- **Phase 2:** LSB Steganography (Statistical hiding).
- **Phase 3:** Anti-Forensic Evasion (Using AI to mimic noise).
- **Phase 4 (The Zheng Era): Hardware-Software Synergy.** The field has moved from analyzing the *bits* (which can be forged) to analyzing the *relationship between the bits and the physical sensor* that created them. This shift is driven by the need to combat "Deepfakes," where visual inspection is no longer sufficient.

4.)A Steganographic Algorithm for Hiding Data in PDF Files Based on Equivalent Transformation

1. Comprehensive Taxonomy Integration

This paper expands your taxonomy into the realm of **Format-Specific Structural Steganography**.

- **Methodology (Equivalent Transformation):**
  - **Seriation-Based Hiding:** Rearranging the order of entries/objects within the PDF. Since the PDF reader doesn't care about the order, the visual "page display" remains identical.
  - **Feature-Based (Old Methods):** Methods the authors criticize, such as varying line/word spacing or character features, which are easier to detect via visual or statistical analysis.

- **Application Domain:**
    - **Secure Document Exchange:** Using mainstream business formats (PDF) as a covert channel.
    - **Integrity-Preserving Forgery:** A method that allows data hiding without changing the "File Size" or "Hash" of the visual content.

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2. Comparative Technical Analysis

The authors provide a direct comparison between their "Equivalent Transform" method and existing software like **wbStego4.3**.

Metric	**Traditional Spacing Methods**	**wbStego4.3 (Object Hiding)**	**Proposed Seriation Method**
Perceptual Transparency	Low (Disturbs page display)	Medium	**High (No visual change)**
File Size Impact	Increases size	Increases size	**Zero (No new data written)**
Capacity	Low	Low (~2 bytes per object)	**High (Factorial of entries)**
Security/Evasion	Weak against statistics	Weak	**Strong (Anti-statistical)**
Integrity Check	N/A	N/A	**SHA-1 + Chaotic Map**

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3. Critical Gap Identification

The paper identifies several "Gaps" in the forensic literature regarding document integrity:

- **The "Structural Invariance" Gap:** Forensic tools often look for *added* data or *modified* pixels. This paper highlights a gap where information can be hidden by simply changing the **metadata sequence** of internal objects, which most standard forensic tools ignore.
- **The Kerckhoffs Rule Violation:** The authors note that many existing hiding tools rely on "security by obscurity." They fill this gap by using a **SHA-1 generated key** to drive a chaotic map, ensuring that even if the algorithm is known, the hidden data remains secure.

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4. Evolution of Solutions: Narrative Flow

This paper represents a midpoint in the evolution of **Document-Based Offensive Strategies**:

- **Early Phase:** Simple "White-space" steganography (changing spaces between words). Highly detectable and fragile.
- **Middle Phase (Liu et al., 2008): Logical Structure Manipulation.** Moving away from the "visual" layer and into the "file format" layer. This is where the term **"Equivalent Transformation"** becomes vital—the file is technically different but logically/visually equivalent.
- **Modern Phase:** Modern document forensics now includes "Structure Analysis" to detect if the internal object order of a PDF has been randomized, a direct response to the techniques proposed in this 2008 paper.

5.) A STUDY ON DIGITAL IMAGE FORGERY TECHNIQUE AND ITS DETECTION

. Comprehensive Taxonomy Integration

This paper classifies image forgery based on the **Intent** and the **Physical Manipulation** of the image content.

- **By Intent:**
  - **Innocent Tampering:** Benign adjustments like zooming, contrast/brightness enhancement, or cropping that do not fundamentally change the meaning.
  - **Malicious/Wicked Forgery:** Intentional manipulation to distort the truth, mislead people, or impact legal proceedings.
- **By Methodology (Content Level):**
  - **Image Retouching:** Enhancing or masking certain features (e.g., changing background or lighting) to distract the viewer.
  - **Image Splicing (Photomontage):** Combining parts of two or more images to create a fake composite.
  - **Copy-Move (Cloning):** Copying a region within an image and pasting it elsewhere to hide an object or duplicate an entity.

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2. Comparative Technical Analysis

The paper discusses the forensic reliability of digital images as evidence, contrasting the ease of manipulation with the difficulty of detection.

Feature / Technique	**Image Retouching**	**Image Splicing**	**Copy-Move Forgery**
Complexity	Low (Layman can use free tools)	Medium (Requires blending)	Medium

Forensic Impact	Minimal to Moderate	**High** (Creates fake events)	**High** (Obscures evidence)
Detection Method	Statistical/Luminance analysis	Edge/Inconsistency detection	Correlation/Matching algorithms
Legal Status	Often overlooked	**Strictly inadmissible**	**Strictly inadmissible**

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3. Critical Gap Identification

The authors point out a significant gap in the "Security vs. Accessibility" of digital images:

- **The Layman Accessibility Gap:** Because low-cost and free editing software is widely available, even non-experts can perform sophisticated forgery. Most legal frameworks (like the **IT Act**) are struggling to keep up with the volume of forged evidence produced by non-technical users.
- **The Trust Paradox:** Images are considered the most "reliable and powerful evidence" in court, yet they are now the easiest to falsify. The paper highlights the urgent need for robust identification schemes that can confirm the **originality and legitimacy** of digital images.

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4. Evolution of Solutions: Narrative Flow

This paper fits into the **Early to Middle Phase** of your research narrative:

- **Visual Era:** Forgery was detected by the human eye (obvious inconsistencies).
- **Tool-Driven Era (Thapa et al., 2019):** Forgery became sophisticated enough to require "Identification Schemes" and "Detection Techniques." The focus shifted from *looking* at the image to *analyzing* the underlying pixel data for retouching and splicing artifacts.
- **Metadata Era (Your Theme-12):** The next step beyond Thapa's focus. Once attackers realized pixel analysis could catch them, they moved to the "Steganographic" level—hiding their manipulations within the **Metadata** (EXIF/Timestamps) to ensure the file still appears forensic-grade.

6.)A blind steganalysis-based predictive analytics of numeric image descriptors for digital forensics with Random Forest & SqueezeNet

1. Comprehensive Taxonomy Integration

This paper contributes to the **Automated/Machine Learning Steganalysis** branch of your taxonomy.

- **Methodology (Feature Extraction/Embedding):**

    - **Image Embedding Networks:** Using pre-trained models like **SqueezeNet**, **InceptionV3**, and **VGG16/19** not as classifiers, but as "feature extractors" to turn an image into a vector of numbers.
    - **Numeric Descriptors:** The resulting mathematical representation of the image that reveals patterns invisible to the human eye.
- **Classification Models:**
    - **Random Forest (RF):** An ensemble learner that proved most effective in this study.
    - **Support Vector Machine (SVM) & Multilayer Perceptron (MLP):** Alternative models used for comparative synthesis.

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2. Comparative Technical Analysis

The paper provides a rigorous performance evaluation across different "Embedder + Model" pairs. This is excellent for your **State-of-the-Art (SOTA)** synthesis table.

Embedder (Extractor)	Best ML Model	AUC (Area Under Curve)	Key Technical Insight
SqueezeNet	Random Forest	**0.998**	**Best Ensemble:** High accuracy with fewer parameters/low overhead.
Painters	Random Forest	0.999	High pattern recognition for artistic/complex signals.
VGG16	Random Forest	0.997	Deep layers (16) provide robust feature vectors.
InceptionV3	Random Forest	0.992	Factoring convolutions improve efficiency but slightly lower AUC.

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3. Critical Gap Identification

The authors identify a major gap in existing deep learning research for forensics:

- **The "Classifier vs. Extractor" Gap:** Most studies use Deep Learning (DL) models to do everything (end-to-end). This paper argues that using DL models specifically as **Feature Extractors** (Embedders) and then passing that data to traditional ML models (like Random Forest) is more efficient and provides better "Predictive Analytics" for digital forensics.
- **Computational Overhead:** The study highlights that "SqueezeNet" achieves AlexNet-level accuracy with **50x fewer parameters**, addressing the gap for forensic tools that need to run on standard law enforcement hardware without massive GPU clusters.

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4. Evolution of Solutions: Narrative Flow

This paper illustrates the **Transition to Feature-Space Forensics**:

- **Phase 1 (Statistical):** Looking for bit-flips (e.g., Li Zhi, 2003).
- **Phase 2 (Structural):** Looking for file format changes (e.g., Liu et al., 2008).
- **Phase 3 (AI-Embedding - Akanji et al., 2022):** The current era. Forensic investigators no longer look at the image; they look at the **Numeric Descriptor Vector**. If a Random Forest model identifies a pattern in these vectors that matches "stego-signatures," the image is flagged as a forgery.

7.)An analysis of LSB Based Image Steganography Techniques

1. Comprehensive Taxonomy Integration

This paper provides the structural definitions for your taxonomy's **Information Security Requirements** and **Carrier Types**.

- **Security Pillars for Digital Evidence:**
  - **Confidentiality:** Ensuring only authorized forensic examiners read the data.
  - **Integrity:** Preventing unauthorized modification (The core of Theme-12).
  - **Non-Repudiation:** Ensuring a sender/creator cannot deny the existence of a file or its metadata.
- **Carrier Classification:**
  - **Image Steganography:** Identified as the "most common" method for hiding information, specifically within GIF and PNG formats.
  - **Text & Audio/Video:** Differentiated by their complexity and redundancy levels.

---

2. Comparative Technical Analysis

The authors provide a clear "Vs." analysis that helps you distinguish between **Content Forgery** (Cryptography) and **Existence Forgery** (Steganography).

Feature	**Cryptography (Content Protection)**	**Steganography (Existence Protection)**
Visibility	Known message passing (Visible but unreadable)	Unknown message passing (Hidden existence)

Structure	Alters the structure of the message (Ciphertext)	**Does not alter the structure** of the secret message
Goal	Prevents content discovery	Prevents discovery of communication itself
Detection	Easier to spot (Garbled text/data)	**Difficult to detect** (Looks like a normal image)

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3. Critical Gap Identification

Thangadurai and Devi highlight a specific gap regarding **Hybrid Security Models**:

- **The Single-Layer Vulnerability:** The paper argues that Cryptography alone is insufficient because it "triggers suspicion." If an investigator sees an encrypted EXIF tag, they know it has been tampered with.
- **The Complexity Gap:** The authors identify that while Audio and Video steganography are possible, they are "very complex to use," leading to a research gap where **Image-based LSB techniques** remain the primary vector for practical forgery in digital platforms.

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4. Evolution of Solutions: Narrative Flow

This paper marks the **Integration Phase (2014)** in the evolution of forensic techniques:

- **Pre-2010:** Focus was purely on making encryption stronger.
- **2014 Era (Thangadurai & Devi):** The shift toward **Hybrid Models**. This paper advocates for encrypting the message *first* and then hiding it using LSB steganography. In your survey, this explains why modern "Anti-Forensic" tools are so hard to beat—they aren't just hiding data; they are hiding *encrypted* data that looks like natural image noise.
- **Post-2020:** Leads into the AI-driven detection (Akanji et al., 2022) you reviewed earlier, which was developed specifically to find these hybrid patterns.

7.)Analysis of Methods for Hiding Information in PDF Documents and Opportunities for Their Progress

1. Comprehensive Taxonomy Integration

This paper maps the physical and logical layers of the PDF format specifically for forensic "vulnerability" assessment.

- **PDF Structural Layers (Carrier Zones):**
  - **The Body:** Contains objects (dictionaries, streams, arrays) where data can be hidden via encoding variations (e.g., swapping Hex <2B> for Octal \053).

    - **The Cross-Reference (XRef) Table:** An index of objects. Manipulation here allows for "Equivalent Transformation" without changing the visual file.
    - **Streams:** Often compressed using **FlateDecode**. Data can be hidden within the "noise" of these compressed streams.
- **Technique Classification:**
    - **Encoding Variation:** Using different PDF-supported data types (Boolean, Integer, Real) to represent the same information differently, creating a covert channel.
    - **Operator Manipulation:** Using PDF display operators to adjust text positioning invisibly.

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2. Comparative Technical Analysis

The authors conduct a study of open-source algorithms and compare the difficulty of hiding data in different media types.

Media Type	**Redundancy (Capacity)**	**Detection Difficulty**	**Forensic Risk**
Multimedia (Image/Audio)	High	Moderate (Statistical)	High (Commonly scanned)
Plain Text	Very Low	Low	Low
PDF (Structural)	**Moderate**	**High**	**Low (Often ignored by AV)**

Key Finding: The authors emphasize that while multimedia has more "space" for hiding, PDF files are "interesting and practical" because they are rarely suspected during routine digital forensic scans compared to images.

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3. Critical Gap Identification

Sergeev and Khorev identify a **"Detection Tooling Gap"**:

- **The Espionage Case Study:** By citing the 2010 FBI arrest of Russian spies, they highlight that steganography is no longer just for "financial fraud" but is a tool for **State-Sponsored Counter-Intelligence**.
- **The Ineffective Tools Gap:** They quote the National Institute of Justice (2010), noting that newer encoding techniques are being developed faster than detection tools, rendering current forensic software "ineffective."
- **The Integrated Approach:** Most tools look for *one* type of hiding. This paper argues for an **Integrated Detection Algorithm** that analyzes the Header, Body, and Trailer simultaneously for inconsistencies.

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4. Evolution of Solutions: Narrative Flow

This paper represents the **Modern Forensic Response Phase (2020)**:

- **2008 Era:** Discovery of structural hiding in PDF.
- **2010-2015:** Realization that these techniques are used in global espionage (Russian spy case).
- **2020 (Sergeev & Khorev):** The development of **Integrated Detection Systems**. We are moving away from looking for "hidden text" and moving toward identifying **structural anomalies** in the PDF data types and compression filters.

8.)Anti-Forensic Challenges in Digital Forensics Investigations: An Overview of Techniques and Tools

1. Comprehensive Taxonomy Integration

The paper provides a standardized classification of anti-forensic methodologies that you can use to structure your survey's "Offensive" branch.

- **Data Hiding:** The most common technique, including steganography and altering file extensions to mislead examiners.
- **Artifact Wiping:** Permanently deleting or overwriting specific files or metadata (e.g., clearing NTFS logs) to ensure they cannot be recovered.
- **Trail Obfuscation:** Misleading investigators by "prolonging the investigation process." This includes manipulating timestamps or forging logs to point to a different user or time.
- **Attacks Against Forensic Tools:** Exploiting vulnerabilities in forensic software (like EnCase or FTK) to crash the tool or produce false reports when it encounters a "poisoned" file.

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2. Comparative Technical Analysis

The authors highlight two specific examples that demonstrate the "Complexity vs. Ease" trade-off in anti-forensics.

Technique	Complexity for Attacker	Difficulty for Examiner	Forensic Impact
Altering File Extension	Extremely Low	Low (Signature Analysis catches it)	Misleads initial triage/manual inspection.

Data Hiding (Stego)	High	**Extremely High**	Conceals the existence of communication entirely.
Trail Obfuscation	Moderate	High	Breaks the "Chain of Custody" and timeline.
Artifact Wiping	Moderate	**Maximum** (If overwritten)	Results in "Dead Ends" in the investigation.

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3. Critical Gap Identification

Mohammad et al. identify a "Modern Investigative Gap" caused by the rapid evolution of technology:

- **The Capability Gap:** Hackers are developing new anti-forensic tools faster than forensic experts can update their defense protocols. The paper notes that "newer techniques render current detection tools ineffective."
- **The Legal Acceptability Gap:** The primary goal of anti-forensics is to make evidence "legally unacceptable." By manipulating the integrity of a file, the attacker ensures it cannot be used in a "court of law" because its "accurate, reliable, and repeatable" nature is compromised.

9.) Anti-Forensic Challenges in Digital Forensics Investigations: An Overview of Techniques and Tools

1. Comprehensive Taxonomy Integration

This paper provides the authoritative classification for the "Offensive" branch of your taxonomy, specifically targeting how criminals bypass forensic discovery.

- **Data Hiding (The Theme-12 Core):** Identified as the most common anti-forensic technique. It includes steganography and simple extension manipulation (e.g., changing a .jpg to .txt) to evade automated triage tools.
- **Artifact Wiping:** The permanent removal of digital traces (e.g., using software to overwrite slack space or clearing NTFS logs) so that evidence cannot be retrieved even with professional forensic tools.
- **Trail Obfuscation:** The act of misleading investigators by creating "false flags." This involves manipulating timestamps (timestomping) or forging log entries to redirect the investigation toward an innocent party.
- **Attacks Against Forensic Tools:** A highly advanced technique where "poisoned" files are created to crash or exploit vulnerabilities in forensic software like EnCase or FTK, thereby invalidating the entire investigation.

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2. Comparative Technical Analysis

The authors provide a breakdown of the "Complexity vs. Impact" of these techniques, which is vital for your Part A synthesis table.

Anti-Forensic Technique	Implementation Complexity	Forensic Detection Difficulty	Primary Objective
Extension Alteration	Low	Low (Signature Analysis)	Mislead initial manual search.
Data Hiding (Stego)	High	**Extremely High**	Conceal existence of data.
Artifact Wiping	Moderate	**Maximum** (If overwritten)	Destroy evidence integrity.
Trail Obfuscation	Moderate	High	Break the investigation timeline.

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3. Critical Gap Identification

Mohammad et al. identify a significant **"Arms Race Gap"** in current digital forensics:

- **The Tooling Ineffectiveness Gap:** The paper quotes the National Institute of Justice to state that anti-forensic encoding is evolving faster than detection protocols. "Current detection tools are rendered ineffective" by multi-layered obfuscation.
- **The Legal Admissibility Gap:** The paper highlights that the goal of anti-forensics is often not to hide the crime, but to make the evidence **legally unacceptable**. If an investigator cannot prove the "integrity" and "reliability" of the evidence because of metadata manipulation, the evidence is discarded in court.

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4. Evolution of Solutions: Narrative Flow

This paper represents the **Modern "Anti-Forensic" Phase (2024)** of your survey:

- **Early Phase (2000s):** Simple file deletion and renaming.
- **Transition Phase (2010s):** Structural hiding in PDFs (Liu et al., 2008) and LSB image steganography (Thangadurai, 2014).
- **Current Phase (2024 - Mohammad et al.): Active Sabotage.** Moving beyond "hiding" to "misleading and attacking." Forensic investigators must now treat every file as a potential

attack on their own tools and use **Heuristic/AI-based approaches** (like the SqueezeNet model from Akanji et al., 2022) to verify data integrity.

10)Boundary Conditions for the Digital Forensic Use of Electronic Evidence and The Need for Forensic Counter-Analysis

1. Comprehensive Taxonomy Integration

This paper contributes to the **"Legal Admissibility & Regulation"** branch of your taxonomy.

- **The "Token" of Invasion:** Defines the **Search Warrant** as the legal mechanism that permits the state to bypass privacy.
- **Standards of Proof:**
    - **Probable Cause:** A lesser standard of reliability used to *initiate* a search. The paper warns that metadata and network forensics are often used tactically to meet this threshold.
    - **Guilt/Innocence:** A higher standard where the technical integrity of the evidence must be absolute to withstand trial.
- **Constitutional Boundaries:** Focuses on **Privacy, Autonomy, and Dignity** as the three pillars challenged by modern network forensics.

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2. Comparative Technical Analysis: The "Reliability Gap"

Losavio et al. highlight a technical paradox that is central to your comparative synthesis.

Evidence Type	**Legal Purpose**	**Required Reliability**	**Forensic Vulnerability**
Network/System Logs	Securing a Warrant	**Low to Moderate**	Can be spoofed to create "False Probable Cause."
File Metadata (EXIF/Timestamps)	Proving the Case	**High**	Targeted by Anti-Forensics (Theme-12).
Physical Drive Data	Final Conviction	**Maximum**	Subject to "Fruits of the Poisonous Tree" doctrine.

Key Finding: Tactical use of digital data often relies on "system-collected data independent of actual network activity." This means investigators may rely on metadata that hasn't been fully verified for integrity just to get a foot in the door (the search warrant).

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3. Critical Gap Identification

The authors identify a **"Vigilance Gap"** in the judicial oversight of digital evidence:

- **The Expansion of Probable Cause:** The paper warns that as technology develops, the definition of "probable cause" is being stretched, potentially weakening protections against illegal search and seizure.
- **The Need for Counter-Analysis:** This is the most important "gap" for your project. The authors argue that because digital evidence is so powerful, there is a mandatory need for **Forensic Counter-Analysis**—the technical process of trying to *disprove* the integrity of the evidence to ensure it wasn't forged or tampered with.

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4. Evolution of Solutions: Narrative Flow

This paper marks the **Legal Realization Phase (2011)** in your survey's timeline:

- **Pre-2010:** Digital evidence was seen as "magic" and inherently reliable.
- **2011 (Losavio et al.):** The realization that digital evidence is a "tactical tool" that can be manipulated. The focus shifts to **Fourth Amendment** implications and the need to correlate electronic records with physical actions.
- **2024 (Mohammad et al.):** The evolution into the "Anti-Forensic" era where "sabotage" is common, confirming Losavio's early fears about the fragility of digital dignity.

11.)Copyright Protection with EXIF Metadata and Error Control Codes

1. Comprehensive Taxonomy Integration

This paper fits into the **"Proactive Integrity & Authentication"** branch of your taxonomy.

- **The Watermark Source:** Uses specific EXIF tags (Manufacturer, Camera Model, Date/Time Stamp) as the "secret message" to be embedded.
- **Technique (Transform Domain):** Uses **Discrete Cosine Transform (DCT)**. Instead of hiding data in the LSB (which is easily destroyed), it alters selected frequency coefficients, making the "tamper-proof" record part of the image's mathematical structure.
- **Error Control Coding (BCH Codes):** A critical addition. **BCH (Bose-Chaudhuri-Hocquenghem)** codes are used to add redundancy. If an attacker modifies the image, the BCH codes allow the investigator to "recover" the original EXIF data from the damaged file.

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2. Comparative Technical Analysis: The Trinity of Watermarking

The authors define three factors that you should use as metrics in your Part A comparative synthesis table:

Metric	Definition in this Study	Forensic Value
Imperceptibility	Watermarked image quality (High PSNR).	The image must look "original" to avoid suspicion.
Robustness	Survivability after intentional attacks.	Ability to recover EXIF even if the image is cropped or compressed.
Capacity	Number of bits embedded.	How many EXIF tags can be "baked" into the image.

Optimization Strategy: The use of **BCH Codes** is the "optimization" here. It ensures that even if 10-15% of the image data is tampered with, the "true" timestamp and camera model can still be mathematically reconstructed.

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3. Critical Gap Identification

Huang et al. identify the **"Fragility Gap"** in standard metadata:

- **Ease of Editing:** The paper notes that EXIF metadata is "frequently encountered with tampering problems due to the ease of editing digital files."
- **The Volatility Gap:** Standard EXIF is just a header; if you "Save As" a different format or strip the header, the metadata is gone. This paper solves that by moving the metadata **from the header into the pixel data** (DCT coefficients).

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4. Evolution of Solutions: Narrative Flow

This paper marks the **Hardware-Linked Authentication Phase (2008)**:

- **Pre-2005:** EXIF was just for "organizing photos."
- **2008 (Huang et al.):** EXIF becomes a **Legal Tool**. By embedding camera-specific metadata into the DCT domain, the file carries its "Birth Certificate" inside its pixels.

- **2020+ (Modern Era):** This leads directly to the **Hardware PUF (Physical Unclonable Function)** hashes you've seen in newer papers, which automate this "watermarking" at the silicon level of the camera sensor.

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12)Copyright Protection with EXIF Metadata and Error Control Codes

1. Comprehensive Taxonomy Integration

This paper fits into the **"Proactive Integrity & Authentication"** branch of your taxonomy.

- **The Watermark Source:** Uses specific EXIF tags (Manufacturer, Camera Model, Date/Time Stamp) as the "secret message" to be embedded.
- **Technique (Transform Domain):** Uses **Discrete Cosine Transform (DCT)**. Instead of hiding data in the LSB (which is easily destroyed), it alters selected frequency coefficients, making the "tamper-proof" record part of the image's mathematical structure.
- **Error Control Coding (BCH Codes):** A critical addition. **BCH (Bose-Chaudhuri-Hocquenghem)** codes are used to add redundancy. If an attacker modifies the image, the BCH codes allow the investigator to "recover" the original EXIF data from the damaged file.

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2. Comparative Technical Analysis: The Trinity of Watermarking

The authors define three factors that you should use as metrics in your Part A comparative synthesis table:

Metric	Definition in this Study	Forensic Value
Imperceptibility	Watermarked image quality (High PSNR).	The image must look "original" to avoid suspicion.
Robustness	Survivability after intentional attacks.	Ability to recover EXIF even if the image is cropped or compressed.
Capacity	Number of bits embedded.	How many EXIF tags can be "baked" into the image.

Optimization Strategy: The use of **BCH Codes** is the "optimization" here. It ensures that even if 10-15% of the image data is tampered with, the "true" timestamp and camera model can still be mathematically reconstructed.

---

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13.)Cryptography and Computational Approaches in Ensuring Data Integrity for Digital Forensic Evidence

1. Comprehensive Taxonomy Integration

The paper broadens your taxonomy by categorizing **Digital Evidence Sources** and the **Computational Approaches** used to protect them.

- **Evidence Sources:** Moves beyond simple files to include **Volatile Storage**, **Network Traffic**, **Social Networks**, and **IoT devices**.
- **Security Approaches:**
    - **Cryptographic Methods:** Traditional encryption and hashing (e.g., SHA-256).
    - **Trustworthy Third Parties:** Systems that rely on a centralized authority to verify integrity.
    - **Blockchain Technology:** Introduced as a decentralized method to ensure that once evidence is logged, its metadata cannot be altered without detection.

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2. Comparative Technical Analysis: The Integrity evaluation

The authors evaluate security methods based on their "goodness and suitability" for the court of law.

Feature	Cryptography (Traditional)	Blockchain/IoT Integration	Forensic Goal
Integrity Check	Hash comparison.	Distributed Ledger.	Prevent tampering.
Chain of Custody	Manual logging/Records.	Automated, immutable logs.	**Authentication**.
Searchability	Limited to unencrypted data.	Multi-keyword search support.	Efficiency in discovery.
Legal Standing	Widely accepted (Standard).	Emerging (High Admissibility).	Provenance verification.

Key Finding: The paper highlights that while law enforcement has conducted investigations for years, they often lack the "expertise and equipment" to handle the volume of **Cloud-derived evidence**, creating a gap that defense attorneys can exploit.

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3. Critical Gap Identification

Mahajan and Pandit identify several "Forensic Deficiencies":

- **The Expertise Gap:** Law enforcement agencies lack the specialized tools to manage the complexity of multi-platform digital evidence.
- **The Privacy-Integrity Paradox:** Protecting the privacy of a user (via encryption) often makes it difficult for forensics to verify the integrity of the hidden data without a "Trustworthy Third Party."
- **The Admissibility Challenge:** In court, if a plaintiff cannot prove a "proper chain of custody" (recording every hand that touched the evidence), the digital evidence—no matter how technically sound—is deemed **inadmissible**.

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4. Evolution of Solutions: Narrative Flow

This paper represents the **Modern Integration Phase (2024)**:

- **2008-2014 Era:** Focus on LSB steganography and simple EXIF watermarking.
- **2020 Era:** Use of ML and SqueezeNet for detection (Akanji et al., 2022).
- **2024 (Mahajan & Pandit):** The shift toward **Blockchain and IoT**. The solution is no longer just "finding" tampered metadata, but building a system where metadata integrity is **guaranteed by design** through distributed computational approaches.

14.) deep learning based steganalysis against spatial domain steganography

1. Comprehensive Taxonomy Integration

The paper defines the structural layers of an AI-driven steganalysis model, which you can use to taxonomize "Detection Methodologies."

- **Pre-Processing Layer:** Uses **5 High-Pass Filters**. These are designed to suppress the image content and highlight the "noise"—which is where the LSB steganographic message resides.
- **Feature Extraction (Convolutional Layers):** 5 layers that automatically learn the statistical patterns of steganography without human intervention.
- **Classification (Fully Connected Layers):** 3 layers that decide if an image is "Original" or "Stego."

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2. Comparative Technical Analysis

Nishimura compares traditional human-led steganalysis with their Deep Learning (DL) approach.

Feature	**Traditional Steganalysis**	**Nishimura's DL Model (2017)**	**Forensic Benefit**
Human Role	Define flaws & statistical anomalies.	None (End-to-end learning).	Removes human bias/error.
Accuracy (LSB)	~80% (Depends on payload).	**91% - 93%** (for BOSSBase).	Higher reliability for evidence.
Domain	Spatial Domain only.	Spatial Domain (with potential for others).	Detects pixel-level tampering.
Efficiency	O(n) based on manual rules.	O(n log n) during training.	Faster detection once trained.

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3. Critical Gap Identification

The authors identify two major gaps in forensic science at the time:

- **The "Human Bottleneck":** Previous methods required experts to manually determine which "filters" to use. This model automates the discovery of steganographic signatures.
- **Universal Steganalysis Gap:** While they achieved high accuracy, the authors admit that a "Universal" method (one AI that catches *all* types of steganography) is still a research requirement.

4. Evolution of Solutions: Narrative Flow

This paper represents the **Automation & AI Phase (2017)**:

- **2008-2014:** Focus on manual LSB and EXIF structural manipulation.
- **2017 (Nishimura):** The introduction of **CNNs (Convolutional Neural Networks)**. Instead of looking for metadata changes, the AI looks for the "shadows" left by the data hiding process in the pixels.
- **2022+:** Leads to the **SqueezeNet/MobileNet** optimizations (Akanji et al.) you reviewed, which made these AI models small enough to run on a forensic investigator's laptop or mobile device.

15.)Deep Learning Based Image Steganalysis

1. Comprehensive Taxonomy Integration

The paper categorizes steganalysis methodologies based on the degree of human involvement and the specific architectural layers of the detection model.

- **Detection Methodologies**:
  - **Traditional Steganalysis**: Relies on human intervention to determine steganography flaws and design specific detection rules.
  - **Deep Learning-based Steganalysis**: An automated approach that defines features or patterns from big data without manual rule-setting.
- **Structural Layer Classification**:
  - **Pre-Processing Layer**: Utilizes a **High-Pass Filter (HPF)** to suppress image content and isolate the "noise" where LSB messages reside.
  - **Feature Extraction (Convolutional Layers)**: A 2-layer sequence (Convolution, ReLU activation, and Local Response Normalization) that automatically learns statistical patterns.
  - **Classification (Fully Connected Layers)**: A 2-layer architecture culminating in a **SoftMax function** to decide if an image is "Original" or "Stego."

2. Comparative Technical Analysis

The following table compares traditional human-led steganalysis with the proposed Deep Learning (DL) model based on metrics identified in the research.

Feature	Traditional Steganalysis	Proposed DL Model (2017)	Forensic Benefit
Human Role	Experts must manually define flaws/anomalies.	None; end-to-end automated learning.	Removes human bias and error.
Accuracy (LSB)	~80% (Variable based on payload/method).	**98%** (Same key) and **90%** (Different keys).	Higher reliability for digital evidence.
Domain	Spatial domain focus.	Spatial domain (LSB-specific).	Detects pixel-level tampering.
Efficiency	Manual rule-based $O(n)$.	Automated once trained.	Faster detection in large datasets.

3. Critical Gap Identification

The authors identify two primary limitations in the state of forensic science regarding hidden message detection:

- **The "Human Bottleneck"**: Previous steganalysis required researchers to manually expose steganography flaws to design a tool. This model addresses the gap by automating feature discovery.
- **Universal Steganalysis Gap**: A single method capable of detecting all types of steganography (spatial and frequency domains) is not yet feasible. The authors admit that while their target-specific model is highly accurate for LSB, a "Universal" AI remains a research requirement.

4. Evolution of Solutions: Narrative Flow

The field of steganalysis has evolved from manual, expert-driven inspection to automated, deep-learning ecosystems.

1. **Manual Detection Phase**: Early steganalysis depended entirely on human researchers identifying statistical flaws in specific embedding methods.
2. **Target-Specific Automation**: The field shifted toward designing algorithms specific to certain domains (like spatial domain LSB) but still required manual feature engineering.

3. **The Deep Learning Phase (2017)**: As shown in this paper, the transition to **Convolutional Neural Networks (CNNs)** allows the system to learn the "noise" of hidden messages automatically.
4. **Future Convergence**: The evolution is moving toward increasing the depth of neural networks and tuning activation functions to move closer to **Universal Steganalysis**.

16) Deep Learning Based Image Steganalysis,

Based on the systematic review provided, here is the comprehensive analysis of deep learning applied to steganalysis, following the requested structural framework.

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1. Comprehensive Taxonomy Integration

The research classifies the evolution of steganalysis by the transition from human-engineered features to automated, end-to-end deep learning architectures.

- **Traditional Steganalysis (Two-Stage)**:
  - **Feature Extraction**: Manual extraction of sophisticated features, primarily using **Rich Models (RM)**.
  - **Classification**: Separate binary classification using **Ensemble Classifiers (EC)**, Support Vector Machines (SVM), or perceptrons.
- **Deep Learning Steganalysis (One-Step)**:
  - **Unified Architecture**: Integrates feature extraction and classification into a single network where parameters are optimized simultaneously.
  - **Domain Applications**: Categorized into the **Spatial Domain** (direct pixel modification, e.g., LSB) and the **Frequency Domain** (JPEG coefficient modification).
  - **Learning Paradigms**: Includes supervised learning (most common), unsupervised learning (early approaches like Auto-Encoders), and Generative Adversarial Networks (GANs).

2. Comparative Technical Analysis

The following table compares the performance and characteristics of the most significant Convolutional Neural Networks (CNNs) identified in the literature against traditional methods.

Model / Method	Year	Key Architectural Innovation	Error Rate (WOW 0.4 bpp)	Error Rate (S-UNIWARD 0.4 bpp)
SRM + EC	2012	Manual Rich Model features + Ensemble Classifier.	25.5%	24.7%
QianNet	2015	First supervised CNN; uses Gaussian activation.	29.3%	30.9%
XuNet	2016	Absolute Value Layer (ABS) and Batch Normalization (BN).	20.7%	27.2%
YeNet	2017	SRM-based filter initialization + Truncation Linear Unit (TLU).	23.2%	31.2%
Yedroudj-Net	2018	Unifies XuNet and YeNet features; uses extended databases.	14.1%	22.8%
ZhuNet	2018	Separable convolutions + Spatial Pyramid Pooling (SPP).	**11.8%**	**15.3%**

3. Critical Gap Identification

Despite the promising results of deep learning, several gaps remain in the current literature:

- **The "Trick" Dependency**: Many high-performing CNNs still rely on "tricks" or heuristics, such as initializing filters with SRM values rather than learning them purely from scratch.
- **Cover-Source Mismatch**: There is a significant performance drop when a network is trained on one database (e.g., BOSSBase) and tested on a different one, highlighting a lack of generalization.
- **Computational Cost for High Resolution**: Most experiments are limited to 256×256 images due to the massive GPU memory and processing power required for high-resolution forensic analysis.
- **Quantitative Steganalysis**: Most current research focuses on binary detection (stego vs. cover) rather than predicting the exact **payload** size embedded in an image.

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4. Evolution of Solutions: Narrative Flow

The field has progressed through four distinct phases of technological sophistication:

1. **Foundational Phase (2014)**: Started with unsupervised learning using **Stacked Auto-Encoders**. Results were initially inferior to traditional Rich Models but proved that neural networks could extract relevant features.
2. **Supervised Refinement (2015–2016)**: The introduction of **QianNet** and **XuNet** brought supervised learning and domain-specific layers like **ABS** to capture noise symmetry.
3. **Hybridization & Depth (2017)**: Researchers began incorporating domain knowledge (SRM filters) into the network initialization and explored much deeper architectures, such as **ResNet-inspired** models with 20+ layers.
4. **Advanced Optimization (2018–Present)**: Current solutions like **ZhuNet** utilize advanced computer vision techniques—**Separable Convolutions** and **Spatial Pyramid Pooling**—to significantly outperform traditional methods and handle arbitrary image sizes.

17) Deep learning based steganograhy

1. Comprehensive Taxonomy Integration

The literature categorizes the evolution of steganalysis from manual, expert-driven methods to automated, data-driven deep learning architectures.

- **Detection Methodologies**:
    - **Traditional Steganalysis**: Relies on human intervention to manually extract features and define statistical anomalies or flaws in steganography.
    - **Deep Learning-based Steganalysis**: An end-to-end approach that unifies feature extraction and classification into a single architecture, learning patterns directly from data without human intervention.

- **Media and Domain Classification**:
    - **Spatial Domain**: Messages are embedded at the pixel level, often by modifying the Least Significant Bits (LSB).
    - **Frequency (Transform) Domain**: Messages are hidden in coefficients after transformations such as Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT), common in JPEG images.
- **AI Architecture Layers**:
    - **Pre-processing**: Uses High-Pass Filters (HPF) to suppress image content and highlight steganographic "noise".
    - **Feature Extraction**: Convolutional layers that automatically learn statistical patterns.
    - **Classification**: Fully connected (Dense) layers that finalize the decision between "Original" and "Stego".

2. Comparative Technical Analysis

The most advanced model discussed across the texts is **SRNet** (Steganalysis Residual Network), which minimizes manual heuristics. The following table compares three independent implementations of SRNet based on their performance metrics at various payloads (Bytes Per Pixel, or bpp).

Performance Metrics for SRNet Implementations

Implementation	Training Epochs	Payload (0.4 bpp) Accuracy	Payload (0.4 bpp) Precision	Error Probability (Per)
Luhanko (Own)	80	0.7443	0.7844	0.1279
TensorFlow	200	**0.8335**	**0.874**	**0.0833**
PyTorch	500	0.7539	0.775	0.1231

Note: The TensorFlow implementation showed the highest efficiency. This is attributed to an additionally introduced dense layer (512 nodes), which improved classification accuracy.

3. Critical Gap Identification

Despite the high accuracy of current models, researchers identify several persistent challenges in the field:

- **The "Human Bottleneck"**: While deep learning automates much of the process, older methods still require experts to manually determine which statistical filters to use.
- **Universal Steganalysis Gap**: A single "Universal" method that can detect all types of steganography across all domains remains a significant research requirement.
- **Hyperparameter Sensitivity**: Designing effective detectors requires intensive fine-tuning of hyperparameters and architecture optimization.
- **Cover-Source Mismatch**: Performance can drop significantly when a model is trained on one database and tested on another with different capture characteristics.

4. Evolution of Solutions: Narrative Flow

The field has transitioned from manual rule-based systems to highly complex residual networks.

1. **Manual/Traditional Phase**: Experts identified flaws in steganography to design specific rules. Accuracy was often limited to ~80% and was highly dependent on the payload.
2. **Early Automation (2014-2015)**: The first CNNs were introduced, such as **QianNet** (2015), which used supervised learning and 5 convolutional layers to achieve results competitive with traditional models.
3. **Refined Architectures (2016-2017)**: Models like **XuNet** (2016) introduced Absolute Value layers (ABS) and Batch Normalization (BN) to improve statistical modeling. **YeNet** (2017) further improved performance by using filter banks initialized with values from traditional Rich Models.
4. **The Residual Phase (2018-Present)**: **SRNet** and **ZhuNet** represent the current state-of-the-art. These models use deep residual connections (20+ layers) to avoid the "gradient vanishing" phenomenon, allowing for much deeper networks and higher accuracy in detecting modern steganographic algorithms like S-UNIWARD and WOW.

18) Deep Learning Hierarchical Representations

1. Comprehensive Taxonomy Integration

Steganalysis methodologies are classified by their reliance on human expertise and the specific domain of the digital image being analyzed.

- **Detection Methodologies**:
  - **Traditional (Two-Stage)**: Requires experts to manually define steganographic flaws and use **Spatial Rich Models (SRM)** for feature extraction, followed by an **Ensemble Classifier (EC)** or SVM.
  - **Deep Learning (Unified)**: Automates both feature extraction and classification within a single CNN framework.
- **Media and Application Domains**:

    - **Spatial Domain**: Detects modifications in the pixel values (e.g., LSB, HUGO, S-UNIWARD).
    - **Frequency (JPEG) Domain**: Targets modifications in transform coefficients, such as DCT or DWT (e.g., J-UNIWARD).
- **Architectural Components**:
    - **Pre-processing Layer**: Employs High-Pass Filters (HPF) to suppress image content and isolate steganographic noise.
    - **Hierarchical Feature Extraction**: Successive convolutional layers that learn statistical artifacts at multiple scales.
    - **Classification Module**: Fully connected layers concluding with a **Softmax** function to differentiate between "cover" and "stego" images.

---

2. Comparative Technical Analysis

The research highlights how different CNN architectures have refined detection accuracy, particularly against adaptive steganography.

Feature	**Traditional SRM + EC**	**QianNet (2015)**	**Xu-Net (2016)**	**YeNet (2017)**	**SRNet (2018)**
Feature Extraction	Manual (Rich Models)	Automated CNN	Automated + ABS layer	30 SRM-based HPFs	Deep Residual Learning
Activation	N/A	Gaussian Function	TanH / ReLU	Truncated Linear Unit (TLU)	ReLU / Clipped ReLU
Accuracy (approx.)	Baseline	~69% (S-UNIWARD)	Outperformed SRM	>90% (Low rates)	State-of-the-Art
Training Focus	Manual rules	Shallow architecture	Spatial/JPEG domain	Spatial domain only	Both Spatial and JPEG

---

3. Critical Gap Identification

Researchers identify several limitations in the current state of forensic science for image steganalysis:

- **The Heuristic Dependency**: Many top-tier networks (e.g., YeNet) still rely on "tricks" like initializing pre-processing filters with SRM values rather than learning them purely from data.

- **Universal Steganalysis Gap**: Most models are specialized for one domain (Spatial or JPEG). A truly "Universal" AI that catches all steganographic types without domain-specific tuning is still a research requirement.
- **Adversarial Vulnerability**: Like many deep learning models, CNN-based steganalyzers are vulnerable to adversarial perturbations (adversarial stego images) that can evade detection.
- **The "Human Bottleneck"**: While feature extraction is automated, the initial selection of "filters" or pre-processing steps often remains human-led.

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4. Evolution of Solutions: Narrative Flow

The field has evolved through four key technological phases:

1. **Manual/Rich Model Phase (Pre-2014)**: Dominated by Spatial Rich Models (SRM) and Ensemble Classifiers. Detection relied on manual exposure of steganography flaws.
2. **Initial Automation (2014–2015)**: The first CNNs (e.g., Tan & Li, QianNet) unified extraction and classification. These early models approached traditional performance but struggled with convergence.
3. **Refined Pre-processing (2016–2017)**: Introduction of the **ABS layer** (Xu-Net) and the **Truncated Linear Unit (TLU)** activation (YeNet) allowed networks to better extract weak steganographic signals from complex image textures.
4. **Deep Residual and Resilient Phase (2018–Present)**: Models like **SRNet** and **ZhuNet** utilized deep residual connections to prevent gradient degradation. Recent work focuses on robustness against image transformations like resizing or noise.

19) Deep learning for steganographic

1. Comprehensive Taxonomy Integration

The literature categorizes steganalysis by the transition from expert-designed manual features to automated, hierarchical feature learning.

- **Detection Methodologies**:
  - **Traditional (Two-Step)**: Relies on human intervention to extract features via **Rich Models (RM)** and classify them using **Ensemble Classifiers (EC)** or SVMs.
  - **Deep Learning (Unified)**: Integrates feature extraction and classification into a single end-to-end architecture, reducing the complexity of manual engineering.
- **Application Domains**:
  - **Spatial Domain**: Targets modifications in the pixel level, such as **LSB** or adaptive algorithms like **S-UNIWARD** and **WOW**.
  - **Frequency (JPEG) Domain**: Targets modifications in DCT coefficients, often requiring "phase-aware" or hybrid architectures.
- **Structural Layer Classification**:
  - **Pre-processing Layer**: Uses fixed or trainable **High-Pass Filters (HPF)** to suppress image content and amplify weak steganographic noise.
  - **Feature Extraction**: Convolutional layers that learn statistical patterns without manual rule-setting.
  - **Classification**: Fully connected layers concluding with a **Softmax function** to provide a probability of the image being "Cover" or "Stego".

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2. Comparative Technical Analysis

The research highlights a clear performance gap between traditional methods and evolving deep learning architectures.

Feature	**Traditional Steganalysis**	**Early CNNs (e.g., QianNet)**	**Modern Residual Nets (SRNet/ZhuNet)**
Human Role	Expert-defined flaws and anomalies.	Manual HPF selection but automated features.	Minimal heuristics; end-to-end learning.
Architectural Key	**SRM + EC**.	5 Convolutional layers + Gaussian Activation.	20+ layers with **Skip-connections** and **BN**.
Accuracy (LSB 0.4)	~80% (Payload dependent).	Competitive with SRM.	**98%** (Same key) / Significantly higher reliability.
Efficiency	Manual rule-based $O(n)$.	High training cost; fast detection.	Automated; handles arbitrary image sizes via **SPP**.

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3. Critical Gap Identification

Researchers identify several major gaps in current digital forensic capabilities:

- **The "Human Bottleneck"**: Early CNNs still required experts to decide which "filters" to use for pre-processing. While models like **SRNet** reduce this, automating the *discovery* of these filters is a continuing goal.
- **Universal Steganalysis Gap**: Most models are "target-specific" (e.g., LSB-only). An AI that catches *all* types of steganography (spatial and frequency) simultaneously remains a research challenge.
- **Cover-Source Mismatch**: Networks often perform poorly when trained on one database (e.g., BOSSBase) and tested on another with different capture characteristics.
- **Adversarial Vulnerability**: Emerging research shows that steganographers can use **Generative Adversarial Networks (GANs)** to create "adversarial stego images" specifically designed to fool CNN detectors.

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4. Evolution of Solutions: Narrative Flow

The field has moved chronologically from simple cryptocurrency-era detection to complex AI ecosystems:

1. **Manual Phase (Pre-2014)**: Relied entirely on expert-led statistical analysis and Rich Models.
2. **Initial Automation (2015-2016)**: The first supervised CNNs (QianNet, XuNet) introduced **ABS layers** and **Batch Normalization** to capture noise symmetry, but still lagged behind SRM in some adaptive scenarios.
3. **Hybridization (2017)**: Models like **YeNet** began using **SRM-inspired initialization** for filters and **Truncated Linear Units (TLU)** to increase the Signal-to-Noise Ratio.
4. **Deep Residual Phase (2018-Present)**: Current state-of-the-art models like **SRNet** utilize deep residual learning to prevent gradient degradation, allowing for 20+ layers that outperform traditional methods across both spatial and JPEG domains.

20) developing trends and challenges

1. Comprehensive Taxonomy Integration

The literature categorizes steganalysis by its structural layers and its evolution from human-led statistical analysis to automated feature learning.

- **Detection Methodologies**:
  - **Traditional (Two-Step)**: Requires experts to manually define steganographic flaws and use **Spatial Rich Models (SRM)** for feature extraction, followed by an **Ensemble Classifier (EC)** or SVM.
  - **Deep Learning (Unified)**: Integrates feature extraction and classification into a single end-to-end architecture, where parameters are optimized simultaneously.
- **Media and Application Domains**:
  - **Spatial Domain**: Targets modifications at the pixel level, such as **LSB** or adaptive algorithms like **S-UNIWARD** and **WOW**.
  - **Frequency (JPEG) Domain**: Targets modifications in transform coefficients (e.g., **J-UNIWARD**).
- **Architectural Components**:
  - **Pre-processing Layer**: Employs High-Pass Filters (HPF) to suppress image content and isolate the "noise" where hidden messages reside.
  - **Feature Extraction**: Convolutional layers that learn hierarchical patterns directly from raw data.
  - **Classification Module**: Fully connected layers concluding with a **Softmax** function to differentiate between "cover" and "stego" images.

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2. Comparative Technical Analysis

The research highlights how different CNN architectures have refined detection accuracy and reduced the need for manual heuristics.

Feature	Traditional SRM + EC	Early CNNs (QianNet)	Modern Residual Nets (SRNet)
Feature Extraction	Manual (Rich Models)	Automated CNN	Deep Residual Learning
Accuracy (LSB 0.4)	Baseline (~80%)	Competitive with SRM	**98%** (High reliability)
Human Role	Define flaws/anomalies	Manual HPF selection	Minimal (End-to-end)
Efficiency	Manual rule-based $O(n)$	Faster once trained	Handles arbitrary sizes via **SPP**

---

3. Critical Gap Identification

Researchers identify several major forensic gaps that modern deep learning attempts to address:

- **The "Human Bottleneck"**: Previous methods required experts to manually determine which "filters" to use; while models like **SRNet** automate this, many high-performing networks still rely on SRM-inspired filter initialization.
- **Universal Steganalysis Gap**: Most models are specialized for one domain; a truly "Universal" method that catches *all* types of steganography remains a research requirement.
- **Cover-Source Mismatch**: Networks often fail when trained on one database and tested on another with different capture characteristics.
- **Big Data Challenges**: The massive volume and variety of multimedia data in modern investigations require automated tools that can scale beyond manual forensic inspection.

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4. Evolution of Solutions: Narrative Flow

The field has moved from simple statistical detection to complex, smart contract-level AI ecosystems:

1. **Manual/Rich Model Phase (Pre-2014)**: Relied entirely on expert-led statistical analysis and manual exposure of steganography flaws.

2. **Initial Automation (2015-2016)**: The first supervised CNNs (QianNet, XuNet) unified extraction and classification but still used manual HPFs.
3. **Hybridization (2017)**: Models like **YeNet** incorporated domain knowledge (SRM filter values) into the network initialization to improve results.
4. **Deep Residual Phase (2018-Present)**: State-of-the-art models like **SRNet** and **ZhuNet** use deep residual connections (20+ layers) to outperform traditional methods across spatial and JPEG domains.

21) Digital Forensic Readiness Information System For EJBCA Digital Signature Web Serve

1. Comprehensive Taxonomy

The research classifies the forensic readiness system by its **investigation lifecycle phases** and the **source of digital evidence** used to monitor the EJBCA (Enterprise Java Beans Certificate Authority) environment.

Forensic Investigation Lifecycle Phases

- **Data Collection**: Gathering raw data from diverse system logs.
- **Data Examination**: Parsing and decomposing collected data into a structured database.
- **Data Analysis**: Sorting data by exploit type and matching it to mitigation recommendations.
- **Findings Report**: Visualizing statistics and generating physical reports.

Evidence Source Classification

- **Application Level**: EJBCA Application Logs (User transaction records, input/output data).
- **Network Level**: Snort IDS Logs (Network traffic, IP addresses, packet data).
- **Operating System Level**: Server Logs via Ubuntu Terminal (TLS status, component update status, module encryption).

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2. Comparative Analysis

The study evaluated three alternative concepts for evidence collection based on their ability to capture specific parameters necessary for a forensic investigation.

Technical Comparison of Evidence Parameters

Evidence Parameter	EJBCA App Logs	Snort IDS Tools	Server Logs (Terminal)
Timestamp / Event Date	Yes	Yes	Yes

Log Level / Event Module	Yes	No	No
Network Traffic / IP Address	Yes	Yes	Yes
Attack Alert Messages / Priority	No	Yes	No
TLS Version / Handshake Status	No	No	Yes
Module Version / Update Status	No	No	Yes
Input/Output Packages	Yes	Yes	Yes

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3. Critical Gap Identification

The research identifies several gaps in current EJBCA server management and the limitations of the proposed forensic readiness system:

- **Vulnerability of Web-Based Signatures**: EJBCA digital signatures are inherently vulnerable to OWASP Top 10 web-based attacks, necessitating pre-incident planning rather than just reactive forensics.
- **Detection Gaps**: During performance testing, the system achieved only an **80% success rate** in detecting attack types and providing mitigation recommendations. Two specific attack types from the OWASP Top 10 2021 could not be detected by the current implementation.
- **Complexity and Reporting**: There is a stated need to increase the detection capability for high-complexity attacks and to improve report formatting to make results more understandable for end-users.

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4. Evolution of Solutions

The field's progression is framed from basic digital signature security to integrated forensic readiness systems:

- **Foundation**: Utilization of digital signatures for non-repudiation and authenticity using sign-and-verify algorithms.
- **Problem Shift**: Recognition that even robust PKI systems like EJBCA are susceptible to evolving web vulnerabilities (e.g., OWASP 2021).

- **Integrated Readiness**: Moving from manual log review to an automated **Information System** that integrates the entire forensic lifecycle—collection, visualization, and physical report generation—into a single pipeline.
- **Technical Stack Evolution**: The proposed solution utilizes modern web technologies including **Python (Flask)** for APIs, **Firebase** for cloud database/hosting, and **Flutter** for the frontend interface.

22 Digital image forensics using EXIF data of digital evidence

1. Comprehensive Taxonomy

The research classifies digital image forensics (DIF) and geotagged information into several hierarchical and functional categories based on their application in investigations and data mining.

- **Forensic Process Lifecycle**:
  - **Identification**: Pinpointing relevant sources of digital evidence and documenting materials.
  - **Collection**: Ensuring no modification of evidence; verified via MD5/SHA1 hash values.
  - **Analysis**: Identifying tools and techniques (the most critical phase).
  - **Presentation**: Reporting findings through written reports, graphics, or testimony.
- **Primary DIF Problem Domains**:
  - **Source Identification**: Recognizing the specific device model responsible for the image.
  - **Forgery Detection**: Evaluating file contents to determine if adulteration has occurred.
- **Geotagged Data Categories**:
  - **User-related**: Photo and User ID for portrait analysis.
  - **Temporal**: Time, date, and duration spent at a location.
  - **Geographic**: Spatial distribution via latitude and longitude.
  - **Textual**: Metadata tags, titles, and descriptions.

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2. Comparative Analysis of Forensic Tools

The study provides a technical comparison of different methodologies and software used to extract and analyze digital image evidence.

Feature / Tool	Manual Visual/EXIF Analysis	Google Image Tool +1	PIE / Autopsy / Pic2Map +2
Primary Strength	High reliability in distinguishing specific apps/headers.	High accuracy in place identification and business integration.	Automated visualization and large-set mapping.
Scalability	Low; difficult/time-consuming for large photo sets.	Moderate; limited by privacy protection filters.	High; designed for rapid review of 80+ images.
Privacy Support	Full control; manual review of every metadata description.	Limited; often does not allow the creation of GPS maps due to privacy.	High; features like "hidden from search engines" available in Pic2Map.
Evidence Type	Metadata descriptions and visual content.	Visual content and exact addresses.	GPS metadata, maps, and header robustness.

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3. Critical Gap Identification

The literature highlights several significant challenges and missing elements in the current state of digital image forensics:

- **The "Privacy vs. Utility" Conflict**: While EXIF data is vital for criminologists, it is also a tool for thieves and cyberterrorists. Users often share photos on social media without realizing they are leaking a "digital footprint".
- **The Problem of Manual Scaling**: While manual analysis is more reliable for specific header verification, it is unfeasible for the "vast" amount of data generated by modern IoT and mobile devices.
- **Inconsistent Tool Accuracy**: There is an identified accuracy gap (approx. 50m) between different localization tools, and Google’s automated tools are not always correct.
- **Deepfake and Person Occurrence**: Current standard DIF struggles with the "blind" verification of fake photos where a person's presence must be authenticated through visual context rather than just metadata.

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4. Evolution of Solutions

The field has transitioned from simple file exchange to complex forensic acquisition:

- **Foundational (1992)**: The JPEG File Interchange Format (JFIF) was created purely to enable the exchange of bitstreams between applications.
- **Metadata Integration (1998)**: The EXIF standard was introduced, allowing manufacturers to embed camera settings and timestamps directly into JPEG/TIFF files.
- **Geospatial Layering**: With the integration of GPS micro-chips into smartphones and drones, photos gained a "geotagged layer," turning static images into spatial data points.

- **Automated Forensics**: Solutions moved toward using specific EXIF headers as "digital fingerprints" to identify the acquisition device model.
- **AI and Neural Networks (Future)**: The field is currently moving toward using Convolutional Neural Networks (CNN) for automated face detection and visual content analysis to detect sophisticated forgeries

23 Document Image Forgery Detection Based on Deep Learning Models,

1. Comprehensive Taxonomy

The research classifies image forgery into a hierarchical structure based on the generation method and the subject matter of the image.

- **By Methodology**:
  - **Traditional Forgery**: Uses image editing software for manual manipulation.
    - **Cutting/Splicing**: Stitching together different pictures.
    - **Copying and Moving**: Copying an area within the same image to another location.
    - **Erasing/Removal**: Eliminating an area and filling it with surrounding pixels.
    - **Image Enhancement**: Applying sharpening, blurring, or mosaics.
  - **Deep Forgery**: Uses deep learning algorithms like Autoencoders or Generative Adversarial Networks (GANs).
    - **Face-specific categories**: Complete generation, face replacement, attribute modification, and expression replacement.
- **By Application Domain**:
  - **Ordinary Image Forgery**: General visual content.
  - **Document Image Forgery**: Specifically targeting text-based images, certificates, and scans.

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2. Comparative Analysis

The study compares the proposed model against several existing baseline algorithms using the Area Under the Curve (AUC) metric across different forgery types.

Algorithm Accuracy Comparison (AUC %)

Algorithm	Mixed Forgery Methods	Copying Type	Removal Type
E011	48.1%	48.2%	49.1%
CFA1	49.2%	48.6%	49.4%
J-LSTM	74.6%	72.3%	76.6%
RGB-N	81.5%	80.6%	82.0%
Proposed Model	**82.3%**	**81.8%**	**82.7%**

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3. Critical Gap Identification

The authors identify several limitations in both current literature and their own proposed solution:

- **Lack of Localization**: Most existing detection algorithms only support classification (identifying *if* an image is forged) but cannot determine the specific *location* of the forgery.
- **The "Expert" Bottleneck**: Traditional Error Level Analysis (ELA) requires human experts to manually judge compression inconsistencies, which is subjective and time-consuming.
- **Inability to Categorize Forgery**: While the proposed model can locate forged areas, it currently lacks the capability to distinguish the *type* of forgery used (e.g., whether a located area was copied or deep-forged).
- **Unclear Visual Information**: Standard RGB-based models often fail when forged images leave few visual clues, necessitating the fusion of hidden features like compression levels.

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4. Evolution of Solutions

The narrative of forgery detection has evolved from human-centric observation to multi-modal deep learning architectures.

- **Phase 1: Human Eye Observation**: Early detection relied purely on visual cues, which became unreliable as editing software grew more intelligent.
- **Phase 2: Traditional Forensic Methods**: Discovery of features not visible to the eye, such as JPEG compression artifacts, noise patterns, and EXIF consistency.
- **Phase 3: Deep Learning Backbones**: Adoption of convolutional neural networks (CNNs) like ResNet and EfficientNet to extract hidden layer features more efficiently than manual methods.
- **Phase 4: Multi-Channel Fusion (Current)**: The current state-of-the-art involves fusing traditional forensic signals (like ELA) with modern deep learning features. The proposed model integrates **EfficientNet** for feature extraction, **Local Anomaly Detection** for localization, and **Tesseract OCR** for text-area result optimization.

24 Forensic Detection of Timestamp Manipulation for Digital Forensic Investigation,

1. Comprehensive Taxonomy

The research classifies timestamp manipulation detection into a hierarchy based on the type of system artifacts analyzed.

Artifact-Based Classification

- **File System Artifacts**:
  - **$MFT (Master File Table)**: Analyzes metadata storage rules, entry sequence numbers, and directory index ($INDX) attributes.
  - **NTFS Journals**: Direct analysis of $LogFile (transaction logs) and $UsnJrnl (change logs) to detect behavioral traces.
- **Operating System Artifacts**:
  - **Execution Traces**: Prefetch files and Registry entries to identify the use of manipulation tools.
  - **Linked Artifacts**: LNK files that track when a target file was opened.
  - **System Logs**: Event Logs (user login duration) and Volume Shadow Copies (VSC).

Functional Classification

- **Direct Detection**: Methods that detect the manipulation event itself as it occurs (e.g., $LogFile-1, $UsnJrnl-1).

- **Indirect Detection**: Methods that identify traces or inconsistencies left behind by manipulation (e.g., $MFT-1, Execution-1).

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2. Comparative Analysis

The study compares existing detection methods against key forensic metrics, emphasizing that journal-based methods offer the highest resistance to anti-forensic techniques.

Technical Comparison of Detection Methods

Method Alias	Type	Level	Anti-Forensic Resistance	Primary Limitation
$MFT-1	Indirect	Suspicious	High	Rules can be bypassed if an attacker tampers with all eight timestamps.
$MFT-4	Indirect	Additional	High	High false positives; 100-nanosecond units are often zero in normal cases.
$LogFile-1	Direct	Suspicious	High	Misses manipulations of timestamps other than $SI-C (Creation Time).
Execution-1	Indirect	Additional	Low/Medium	Cannot detect arbitrary or custom malware execution.
VSC-1	Indirect	Suspicious	Low	Fails if file creation and manipulation occur between shadow copy intervals.

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3. Critical Gap Identification

The research identifies several fundamental gaps in previous forensic literature and existing tools:

- **The Tunneling Problem**: Previous NTFS journal-based methods ($LogFile-1, $LogFile-2) frequently misidentify "File System Tunneling"—a legitimate OS function that updates creation times—as malicious manipulation.
- **Detection Range Limitations**: $UsnJrnl-1 cannot detect manipulations performed via the SetFileTime() API or PowerShell because these methods do not change the $SI-E (MFT Entry Modified Time) in a way that triggers traditional detection rules.
- **Fragmented Analysis**: Existing studies focus on individual artifacts in isolation rather than providing an integrated analysis of different detection methods.
- **Anti-Forensic Vulnerability**: Many OS-based artifacts (Prefetch, Event Logs) are easily deleted or disabled by attackers, yet remain a primary focus of many older detection methods.

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4. Evolution of Solutions

The field of file system forensics has evolved from rule-based metadata analysis to behavioral-based journal analysis.

- **Initial Stage (Metadata Rules)**: Early solutions focused on $MFT "normal rules" (e.g., Creation Time should be earlier than Modified Time). Attackers quickly learned to bypass these by manipulating all eight file timestamps simultaneously.
- **Intermediate Stage (OS Artifacts)**: Forensic focus shifted to "side-channel" traces like LNK files and Prefetch execution signatures. This proved ineffective against sophisticated APT (Advanced Persistent Threat) malware that avoids well-known tool names.
- **Advanced Stage (Journal-Based Detection)**: Current state-of-the-art focuses on NTFS journals ($LogFile and $UsnJrnl) because they record the *behavior* of the manipulation itself.
- **Modern Integrated Approach (Proposed Algorithm)**: The latest evolution involves cross-analyzing journals with MFT entries and explicitly identifying legitimate OS behaviors like tunneling to eliminate false positives

25 Forensic Recovery of File System Metadata for Digital Forensic Investigation

1. Comprehensive Taxonomy

The research classifies existing metadata recovery methods based on the **unit of recovery** and the **predictability of data headers**.

Classification by Recovery Unit

- **File-Level Recovery**: Restores the entire $LogFile as a single object.
- **Page-Level Recovery**: Recovers the 4KB units (Restart Pages and Logging Pages) that compose the $LogFile.
- **Record-Level Recovery (Proposed)**: Recovers the individual transaction records within the Logging Pages, which is the most granular and robust form of recovery.

Classification by Data Predictability

- **Fixed-Value Carving**: Methods that rely on "magic numbers" or constants (e.g., "RSTR" for Restart Pages or "RCRD" for Logging Pages) at the start of a data block.
- **Non-Fixed Value Carving (Proposed)**: Methods designed for records that lack consistent headers or range-predictable values at their start position.

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2. Comparative Analysis

The study evaluates the proposed record-level recovery algorithm against established forensic tools, specifically testing performance when metadata structures are damaged.

Forensic Tool Performance Comparison

Metric	Bulk Extractor / Standard Tools	Proposed Record-Level Tool
Recovery Unit	File or Page level only	Individual Record level
Header Dependency	Requires undamaged Page/File headers	Independent of Page/File headers
Data Integrity	Fails if the 4KB page structure is corrupted	Recovers all undamaged internal records

Forensic Utility	Limited event reconstruction	Enables full event and history tracking

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3. Critical Gap Identification

The research identifies a major technical void in current digital forensic literature regarding the New Technology File System (NTFS):

- **The Header Dependency Gap**: Prior research assumes that recovery must start from a fixed value (like a signature). Because $LogFile records do *not* have a fixed value at their start position, no systematic study previously existed for recovering them at the record level.
- **Vulnerability to Corruption**: Existing tools fail completely if the page header of a metadata file is damaged, even if the actual transaction records inside that page are perfectly intact.
- **Standardization Absence**: There was a lack of open-source tools capable of granular, record-by-record carving for NTFS transaction logs, hindering investigators in cases of extreme file system damage.

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4. Evolution of Solutions

The narrative of metadata recovery has moved from recovering simple "objects" to reconstructing the "behavioral history" of the system.

- **Early Evolution (Fixed Carving)**: Solutions were initially developed to "carve" deleted files or metadata based on known signatures (e.g., $MFT or $UsnJrnl headers).
- **Intermediate Phase (Page Recovery)**: Tools like **Bulk Extractor** evolved the field by allowing investigators to recover 4KB pages of $LogFile data using the "RSTR" and "RCRD" fixed values.
- **Advanced Phase (Record-Level Recovery)**: The current breakthrough focuses on the internal structure of these pages. By analyzing how transaction records are stored without fixed headers, the proposed solution allows for the reconstruction of file system operations even when the surrounding structure is destroyed by "cyber terrorism" or physical disasters.

26 "Forgery Attacks on Digital Signature Schemes without using One-way Hash and Message Redundancy

1. Comprehensive Taxonomy

The research classifies digital signature schemes and their vulnerabilities based on their architectural requirements for message integrity.

- **Signature Scheme Architectures**:

    - **Standard Schemes**: Require a one-way hash function and message redundancy padding to resist forgery attacks.
    - **Hash-less Schemes**: Proposed alternative schemes that claim to resist forgery without using one-way hash functions or redundancy.
- **Multi-signature Methodology**:
    - **Parallel Multi-signatures**: Each user signs a document separately and independently; individual signatures are later combined.
    - **Serial Multi-signatures**: A group of signers signs a document in a sequence without a predetermined order.
- **Attack Classification**:
    - **Existential Forgery**: An attack where an adversary can create a valid signature for at least one message, though they may have no control over the message's content.

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2. Comparative Analysis

The study provides a technical critique of the **Chang-Chang scheme** versus conventional cryptographic requirements for mobile code authorization.

Feature	Conventional Schemes	Chang-Chang Scheme
Dependency	Requires One-way Hash Functions	Claims Independence from Hash Functions
Message Padding	Uses Message Redundancy	No Redundancy Required
Efficiency	High computation/communication cost	Higher efficiency for mobile code systems
Security Status	Proven resistant to existential forgery	**Vulnerable** to existential forgery
Integrity Source	Provided by Hash and Redundancy	Provided by modular arithmetic (proven insufficient)

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3. Critical Gap Identification

The research identifies a significant gap between "claimed" efficiency and "actual" security in cryptographic protocol design:

- **The Formal Proof Gap**: The primary gap identified is the lack of **formal security proofs**. The Chang-Chang scheme was designed based on the "interesting" but flawed claim that modular arithmetic alone could replace hash functions.
- **False Sense of Security**: The paper demonstrates that even if a secure one-way function were adopted into the flawed scheme, it would still remain insecure due to the underlying mathematical structure of the protocol.
- **Systemic Vulnerability**: A critical missing element in the targeted literature is the failure to account for how an attacker can manipulate parameters (r, s, y) to satisfy verification equations without knowing the private key.

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4. Evolution of Solutions

The narrative of digital signature evolution in this context moves from complex multi-signer systems to failed attempts at hyper-efficiency.

- **Initial Stage (Authorization Models)**: Development of distributed hypertext systems required secure mobile codes, leading to the creation of novel multi-signature schemes by Shieh et al.
- **Intermediate Stage (Optimization)**: Designers sought to remove "burdensome" components like one-way hash functions to increase the speed and efficiency of computation and communication for mobile devices.
- **The "Hash-less" Proposal**: Chang and Chang proposed a scheme that intentionally omitted hash functions and redundancy, claiming modular exponentiation provided enough security.
- **Current Stage (Cryptanalysis and Refinement)**: Modern research, as presented in this paper, serves as a corrective phase. It re-emphasizes that one-way hash functions are not just "extras" but are fundamental requirements. The field is moving back toward **formal proofs** for all cryptographic protocol designs to prevent existential forgeries.

27 General Countermeasures of Anti-Forensics Categories

. Comprehensive Taxonomy

The research classifies anti-forensics (AF) and their countermeasures into a hierarchical structure based on the **attacker's objective** and the **forensic phase** targeted.

- **By Anti-Forensic Category**:
    - **Data Hiding**: Techniques like steganography or cryptography used to make evidence inaccessible.
    - **Data Deletion**: Permanent removal of evidence using wiping tools or degaussing.
    - **Data Forgery/Trail Obfuscation**: Altering logs, timestamps, or headers to mislead investigators.

    - **Analysis Prevention**: Methods designed to crash forensic tools or exploit known vulnerabilities in software like EnCase or FTK.
- **By Application Domain**:
    - **Network Forensics**: Anti-forensics targeting traffic logs and intrusion detection systems.
    - **Mobile/Cloud Forensics**: Specifically targeting Android/iOS artifacts or distributed data storage.
    - **Media Forensics**: Manipulation of digital image and video compression artifacts.

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2. Comparative Analysis

The study provides a technical comparison of countermeasures against specific anti-forensic threats, highlighting the effectiveness of different investigative approaches.

AF Category	Countermeasure Technique	Primary Metric	Security Impact
Data Hiding	Steganalysis & Statistical Analysis	Detection Accuracy	High; identifies anomalies in file noise/size.
Data Deletion	Journal/Log File Recovery	Data Recoverability	Medium; depends on overwrite frequency.
Trail Obfuscation	Cross-Artifact Correlation	Evidence Integrity	High; identifies inconsistencies across system artifacts.
Analysis Prevention	Sandbox/Virtual Environment Execution	Tool Scalability	High; prevents tool-targeted malware from affecting the host.

---

3. Critical Gap Identification

The research identifies several fundamental "bottlenecks" and missing elements in current digital forensic literature and practice:

- **The "Arms Race" Gap**: There is a significant lag between the development of new anti-forensic tools (which are often automated) and the creation of their corresponding countermeasures (which are often manual or reactive).

- **Lack of Proactive Models**: Most current literature focuses on **post-mortem analysis**. There is a critical lack of **proactive** or real-time digital forensic models, especially for IoT environments.
- **Standardization Deficit**: The paper notes the absence of a universal, standardized framework for evaluating the "anti-forensic resistance" of new software or file systems.
- **Encrypted Cloud Data**: A major identified gap is the inability of standard forensic processes to handle "Data Hiding" when implemented via high-level cloud encryption where keys are not locally stored.

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4. Evolution of Solutions

The field of digital forensics has evolved through distinct thematic stages in response to anti-forensic techniques.

- **Initial Stage (Simple Recovery)**: Early forensics focused on basic data recovery of deleted files using simple carving techniques.
- **Intermediate Stage (Signature Analysis)**: In response to data hiding, the field moved toward **signature-based detection** and hash-value verification (MD5/SHA1) to ensure data integrity.
- **Advanced Stage (Behavioral & Statistical)**: As attackers began using trail obfuscation and encryption, solutions shifted to **statistical steganalysis** and behavioral analysis of system journals and logs.
- **Current/Future Stage (Integrated & Proactive)**: The modern evolution involves **proactive IoT evidence analysis** and integrated frameworks that assume the presence of anti-forensics by default, utilizing AI and chaotic stream ciphers to protect the forensic process itself.

28 Hash-algorithms Output for Digital Evidence in Computer Forensics

1. Comprehensive Taxonomy

The research classifies hash algorithms and their forensic utility based on their **mathematical properties** and their **functional application** within the investigation lifecycle.

- **By Mathematical Property**:
  - **One-Wayness**: The inability to reverse the output string to find the original input.
  - **Collision-Free Nature**: The requirement that two different inputs must not produce the same hash output.
  - **Fixed Output Length**: Transforming any variable-length digital string into a consistent, short-string output.
- **By Forensic Application Phase**:
  - **Collection/Acquisition**: Creating a "fingerprint" of the original evidence at the crime scene.
  - **Authentication/Verification**: Comparing current hash values against the original to prove no minor changes occurred.

- **Storage**: Maintaining the reliability of evidence during the chain of custody.

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2. Comparative Analysis

The study reviews the most prominent hash algorithms used in computer forensics, comparing them based on their bit-length and security characteristics.

Algorithm	Bit Length	Core Security Metric	Forensic Utility
MD5	128-bit	Speed over high security	Widely used but vulnerable to collision attacks.
SHA-1	160-bit	Standard authentication	More secure than MD5; used for evidence verification.
SHA-256	256-bit	High collision resistance	Modern standard for high-integrity requirements.
RIPEMD-160	160-bit	European standard	Alternative to SHA-1 developed in Europe.

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3. Critical Gap Identification

The research identifies a significant technical and procedural challenge in the current forensic landscape:

- **The "Minor Change" Vulnerability**: A critical gap is the sensitivity of hash functions; while they are excellent at detecting *any* change, they cannot distinguish between malicious tampering and benign metadata updates (like a system-generated timestamp change), which can lead to evidence being ruled inadmissible.
- **Hash Cracking Risks**: The paper highlights a gap in the longevity of evidence; as computational power increases, older hash outputs (like MD5) become susceptible to "cracking," potentially compromising the long-term authentication of archived evidence.
- **The Scalability Problem**: As digital storage sizes grow exponentially, the time required to calculate hashes for entire drives during the collection phase creates a significant bottleneck in rapid incident response.

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4. Evolution of Solutions

The field of digital authentication has moved from simple data matching to complex cryptographic verification.

- **Initial Stage (Manual Comparison)**: Early forensics relied on bit-stream copies and manual verification, which was error-prone and unscalable.
- **Intermediate Stage (Standardization of Hashing)**: The adoption of MD5 and SHA-1 as the "gold standard" allowed investigators to provide a mathematical proof of evidence integrity that could be presented in court.
- **The Authentication Crisis**: The discovery of mathematical vulnerabilities in MD5 led to a shift toward more complex, longer-string algorithms like the SHA-2 family.
- **Modern Stage (Integrated Verification)**: Current solutions involve automated forensic suites (like EnCase or FTK) that integrate hashing into every step of the lifecycle, moving from stand-alone hash tools to built-in, real-time authentication procedures in cyberspace.

28 Hybrid Statistical and Deep Feature-Based Detection of Steganography in DCT and LSB Domains

1. Comprehensive Taxonomy

The research classifies steganography and steganalysis based on the **embedding domain** and the **feature extraction methodology** used for detection.

- **By Steganography Embedding Domain**:
  - **Spatial Domain**: Techniques like Least Significant Bit (LSB) that modify pixel values directly.
  - **Transform Domain**: Techniques like Discrete Cosine Transform (DCT) used in JPEG images to hide data in frequency coefficients.
- **By Steganalysis Methodology**:
  - **Statistical Steganalysis**: Using mathematical models (e.g., histograms, Chi-square tests) to find deviations in image statistics.
  - **Deep Learning Steganalysis**: Utilizing Convolutional Neural Networks (CNNs) to automatically learn complex patterns and anomalies.
  - **Hybrid Steganalysis (Proposed)**: Combining both statistical and deep features to improve detection across multiple domains.

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2. Comparative Analysis

The study evaluates the performance of the proposed hybrid model against traditional and deep-learning-only approaches across different image formats.

Feature	Statistical Methods	CNN-only Methods	Proposed Hybrid Model

Detection Domain	Limited to specific domains (e.g., LSB)	Generalizable but lacks statistical precision	Cross-domain (DCT & LSB)
Feature Type	Hand-crafted (Histograms)	Automated (Convolutional)	Combined (Multi-modal)
Computational Efficiency	Very High	Low (due to deep layers)	Balanced (Optimized layers)
Accuracy (Mean)	~70-80%	~85-92%	**96.4%**

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3. Critical Gap Identification

The research identifies several gaps in current steganalysis literature that hinder effective digital investigation:

- **The Single-Domain Limitation**: Most existing tools are "domain-specific," meaning a tool designed for LSB (spatial) often fails to detect data hidden in DCT (transform) domains.
- **Feature Insufficiency**: Deep learning models often miss "fine-grained" statistical anomalies that traditional mathematical tests can easily catch, and vice-versa.
- **Robustness to Compression**: There is a lack of standardized models that remain accurate when images undergo secondary processing like JPEG re-compression.
- **Real-time Detection**: The paper highlights a gap in the ability to process large-scale datasets (like social media streams) in real-time without sacrificing high detection accuracy.

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4. Evolution of Solutions

The narrative of steganography detection has moved from detecting simple pixel changes to analyzing high-dimensional neural patterns.

- **Initial Stage (Visual & Simple Statistical)**: Early steganalysis focused on visual inspections and simple LSB histogram analysis to find "perfectly even" bit distributions.
- **Intermediate Stage (Universal Steganalysis)**: Development of feature-rich models like **SPAM (Subtractive Pixel Adjacency Model)** that look for correlations between neighboring pixels.

- **Deep Learning Era**: The shift to CNNs allowed for "blind steganalysis," where the system does not need to know the embedding algorithm beforehand to identify hidden data.
- **Current Stage (Hybrid Fusion)**: The modern evolution (as proposed in this paper) involves **feature-level fusion**. By merging the mathematical certainty of statistical tests with the pattern-recognition power of deep learning, researchers are achieving near-perfect detection rates even in complex, multi-domain environments.

29 Image Forgery Detection Techniques: Latest Trends and Key Challenges,"

1. Comprehensive Taxonomy

The research classifies image forgery detection into two primary methodologies, further categorized by the specific artifacts or techniques used.

- **Active Forgery Detection**: Requires pre-embedded information to verify authenticity.
    - **Digital Watermarking**: Embedding a signature or pattern into the image during capture.
    - **Digital Signatures**: Using cryptographic hashes and metadata to ensure integrity.
- **Passive (Blind) Forgery Detection**: Analyzes the image without any prior information or embedded data.
    - **Pixel-based**: Detecting cloning (Copy-Move), Splicing, and Resampling (scaling/rotation).
    - **Format-based**: Analyzing JPEG compression artifacts and double-compression traces.
    - **Camera-based**: Identifying sensor noise (PRNU), Color Filter Array (CFA) interpolation, and lens aberration.
    - **Physical-based**: Checking for inconsistencies in lighting, shadows, and reflections.
    - **Geometry-based**: Analyzing the principal point and vanishing points in a 3D scene.

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2. Comparative Analysis

The survey compares traditional (hand-crafted) methods against modern deep learning-based protocols across several technical metrics.

Metric	Traditional (Hand-crafted)	Deep Learning (CNN/GAN)
Scalability	Low; often specific to one attack type.	High; can learn features for multiple attack types.

Security	Vulnerable to simple post-processing (e.g., blurring).	Robust against most post-processing and noise.
Efficiency	High; requires less computational power.	Low; requires high-end GPUs for training and inference.
Generalization	Poor; fails on unknown forgery types.	High; excellent at detecting sophisticated AI-generated forgeries.

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3. Critical Gap Identification

The research identifies several fundamental "bottlenecks" that remain unresolved in current literature:

- **The Localization Gap**: While many systems can determine if an image is forged (classification), pinpointing the *exact* modified pixels (localization) remains highly challenging, especially in complex "seamless" splices.
- **Deepfake Sophistication**: The rapid evolution of Generative Adversarial Networks (GANs) has created a gap where forgery generation techniques often outpace the development of detection tools.
- **Standardized Dataset Shortage**: Current research lacks large-scale, standardized, and diverse datasets that include real-world social media compressions, making it difficult to benchmark different protocols fairly.
- **The Post-Processing Vulnerability**: Many detection algorithms lose significant accuracy when images are subjected to common operations like social media re-compression or minor resizing.

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4. Evolution of Solutions

The field has transitioned from manual, geometry-based verification to fully automated, feature-aware neural systems.

- **Initial Stage (Active Protection)**: Early efforts relied on watermarking and hardware-level signatures. However, these were limited because most cameras did not support them, and users rarely implemented them.
- **Intermediate Stage (Passive Pixel Analysis)**: The focus shifted to "blind" detection, analyzing statistical anomalies in pixels and JPEG compression. Key tools were developed to detect "Copy-Move" and "Splicing" through hand-crafted features like SIFT or SURF.
- **Deep Learning Revolution**: The introduction of Convolutional Neural Networks (CNNs) allowed the field to move away from manual feature engineering. Models began automatically learning "hidden" artifacts left by the camera sensor or editing software.

- **Modern Era (Multi-modal & GAN Forensics)**: Current solutions now combine multiple streams of data (e.g., RGB channels + Noise channels) and use specialized architectures to detect AI-synthesized content (Deepfakes), marking the shift toward "complex Smart Forensics" ecosystems.

30 Image Tampering Detection Using Error Level Analysis and Metadata Analysis,

1. Comprehensive Taxonomy

The research classifies image forgery and its detection into a hierarchy based on the **nature of the forensic approach** and the **specific tampering method**.

- **By Forensic Methodology**:
  - **Active Authentication**: Methods requiring prior information like digital watermarking or signatures.
  - **Passive (Blind) Authentication**: Detecting forgeries without any prior knowledge of the original image.
- **By Tampering Technique**:
  - **Copy-Move (Cloning)**: Copying a part of an image and pasting it elsewhere in the same image.
  - **Splicing**: Combining parts of two or more different images to create a fake composite.
  - **Image Retouching**: Enhancing features (like color or contrast) to improve appearance or mislead the viewer.
  - **Morphing**: Digitally transforming one image into another.

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2. Comparative Analysis

The study evaluates the effectiveness of the proposed hybrid approach—combining **Error Level Analysis (ELA)** and **Metadata Analysis**—against traditional single-method detection.

Metric	**Metadata Analysis Only**	**Error Level Analysis (ELA) Only**	**Proposed Hybrid Model**
Detection Basis	File structure and software headers.	Compression rate inconsistencies.	Multi-modal (Context + Pixel).

Security/Reliability	Low; metadata is easily wiped or spoofed.	Medium; sensitive to re-compression.	High; ELA catches what metadata misses.
Tamper Localization	Cannot pinpoint modified areas.	Identifies specific modified pixel regions.	Pinpoints area and identifies tool used.
Computational Cost	Very Low.	Moderate (requires resaving/differencing).	Balanced for rapid assessment.

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3. Critical Gap Identification

The research identifies several gaps in current literature and existing tools that limit the effectiveness of digital forensics:

- **The "Information Pollution" Challenge**: With the massive volume of digital content, existing literature lacks scalable ways to filter "polluted" (false/misleading) images from legitimate ones in real-time.
- **AI-Driven Forgery Sophistication**: Traditional passive detection methods are increasingly unable to identify "seamless" forgeries created by modern AI and deep learning techniques.
- **Metadata Volatility**: A significant gap is the reliance on metadata; since many social media platforms automatically strip metadata, a system relying solely on it is often useless in real-world investigations.
- **Human Eye Fallibility**: The paper notes that many forged images are now so high-quality that they cannot be reliably identified by the human eye, necessitating automated neural-based verification.

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4. Evolution of Solutions

The field of image forensics has evolved from simple visual checks to complex, multi-layered authentication systems.

- **Initial Stage (Active Protection)**: Early efforts focused on "watermarking," where cameras would embed a hidden code. This failed to scale because it required hardware changes.
- **Intermediate Stage (Passive Forensic Tools)**: Researchers moved toward analyzing the internal "physics" of an image. Tools were developed to check for **JPEG double compression**—the "fingerprint" left when a modified image is resaved.

- **The ELA Era**: Error Level Analysis became a standard for identifying areas of an image with different compression levels, allowing investigators to "see" where objects were added or removed.
- **Modern Hybrid Integration**: The latest evolution (proposed here) integrates **deep neural networks** with ELA and Metadata. Instead of choosing one method, current solutions use metadata for quick screening and ELA for deep pixel-level verification, creating a "complex forensic ecosystem" capable of fighting advanced digital misinformation.

31 LSB Steganography and AES Encryption for Multiple PDF Documents,

1. Comprehensive Taxonomy

The research classifies the protection of digital information based on the **combination of security layers** and the **volume of the carrier media**.

- **Security Layering (Multi-modal)**:
  - **Cryptography**: Using the Advanced Encryption Standard (AES) to transform secret text into unreadable ciphertext before embedding.
  - **Steganography**: Using the Least Significant Bit (LSB) technique to hide the encrypted data within a carrier file so its existence is concealed.
- **Carrier Media Classification**:
  - **Single-Document Steganography**: Traditional methods focusing on hiding data within one file (e.g., one image or one PDF).
  - **Multi-Document Steganography (Proposed)**: Spreading secret information across multiple PDF documents to increase total hiding capacity and reduce the statistical "noise" in any single file.
- **PDF-Specific Methodology**:
  - **Operator-based Hiding**: Specifically utilizing the **Tj Operator** within the PDF structure to embed bits without altering the visual document content.

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2. Comparative Analysis

The study evaluates the proposed **MPDFStego** system against traditional single-document steganography, focusing on capacity and security metrics.

Metric	**Single-Document Steganography**	**MPDFStego (Multi-PDF)**
Capacity	Limited by the size of one carrier file.	Scalable; increases with the number of PDF documents.

Encryption Protocol	Often basic or omitted.	AES-128 in Cipher Block Chaining (CBC) mode.
Detection Risk	High for large payloads (visible artifacts).	Lower; payload is distributed across multiple files.
Tool Dependency	Standalone.	Integrated with QPDF for compression/decompression.

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3. Critical Gap Identification

The research identifies several unresolved issues in the existing literature regarding document-based data hiding:

- **The "Capacity Bottleneck"**: Most existing research has focused on securing a single document. There was a critical lack of attention toward scenarios requiring the transmission of "large information" that exceeds the capacity of a single carrier.
- **Statistical Detectability**: A major gap is that hiding a large amount of data in one file significantly changes its bit-pattern, making it easy for steganalysis tools to detect. Spreading data across multiple files is identified as a necessary but previously under-researched countermeasure.
- **Protocol Integration**: The paper notes a gap in user-friendly programs that automate the entire pipeline of encryption, multi-document splitting, and embedding into specific PDF operators like the Tj Operator.

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4. Evolution of Solutions

The narrative of secure communication has moved from simple data masking to complex, multi-layered "stego-cryptosystems".

- **Initial Stage (Basic Watermarking)**: The field began with putting visible or simple invisible watermarks on documents to claim ownership.
- **Intermediate Stage (Single-File Steganography)**: Development of LSB techniques for images and eventually PDFs, allowing users to hide small text files. However, these were vulnerable if the carrier file was lost or analyzed.
- **Cryptographic Integration**: To address the vulnerability of steganography-only methods, researchers began encrypting data (using AES) before hiding it, ensuring that even if the data was discovered, it remained unreadable.
- **Modern Era (Distributed Multi-Document Systems)**: The current evolution involves moving beyond the single-file limit. Systems like **MPDFStego** represent the shift toward distributed security, where the "secret" is a fragmented, encrypted puzzle spread across a complex ecosystem of multiple digital documents.

32 OleDetection—Forensics and Anti-Forensics of Steganography in OLE2-Formatted Documents

1. Comprehensive Taxonomy

The research classifies steganography within OLE2 documents based on the **specific storage structures** utilized for hiding data.

- **OLE2 Structure-Based Steganography**:
  - **Dead Regions (Unused Space)**: Utilizing "slack" or unallocated space within the OLE2 container that is not currently mapped to a specific stream.
  - **Small Block Depots (SBD)**: Hiding information within the storage units used for small files.
  - **Big Block Depots (BBD)**: Utilizing the primary storage blocks for larger data streams.
- **Forensic Detection Methodology (OleDetection)**:
  - **Step 1: Region Detection**: Identifying all dead or unused regions within the OLE2 document structure.
  - **Step 2: Binary Extraction**: Extracting raw binary data from these identified regions.
  - **Step 3: Statistical Verification**: Applying **kurtosis** and **byte-frequency distribution** analysis to distinguish between random system noise and structured hidden data.

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2. Comparative Analysis

The study evaluates the **OleDetection** algorithm against the earlier **Stegole** algorithm, highlighting technical improvements in detection breadth.

Feature	**Stegole Algorithm**	**OleDetection Algorithm**
Detection Scope	Focused on specific property streams.	Identifies all unused "dead regions" across the document.
Analysis Metric	Basic pattern matching.	Advanced Kurtosis and Byte-Frequency Distribution.
Security/Reliability	Susceptible to missing data in custom streams.	High; statistically detects "structured" binary even without headers.

Application Domain	Microsoft Office OLE2 documents.	Microsoft Office OLE2 documents.

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3. Critical Gap Identification

The research identifies a major conflict between forensic acquisition and evolving document standards:

- **The "Dead Space" Visibility Gap**: Common forensic tools often treat document files as monolithic objects, failing to "look inside" the internal block structures (like BBDs or SBDs) where significant amounts of data can be hidden without changing the document's outward appearance.
- **The Anti-Forensics Consensus Problem**: A major gap identified is the lack of a standardized definition and control mechanism for anti-forensics. This allows criminals to use "stego-channels" in common formats like .doc or .xls to hinder investigations.
- **Statistical Thresholding**: The paper notes a gap in the accuracy of threshold values; while kurtosis helps, distinguishing between a compressed legitimate stream and an encrypted stego-stream remains a significant technical challenge.

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4. Evolution of Solutions

The narrative of document forensics has shifted from analyzing "what is visible" to "what is technically possible" within a file's specifications.

- **Initial Stage (Metadata Analysis)**: Early forensics focused on external metadata (file size, creation dates). Attackers bypassed this by hiding data in "slack space" that didn't affect file size.
- **Intermediate Stage (Stream Analysis)**: Solutions like the **Stegole** algorithm were developed to check known steganography "stego-channels" in document property streams.
- **Structural Forensics (OleDetection)**: The field evolved to treat documents as "virtual file systems." Instead of looking for specific tools, current solutions like **OleDetection** map out the entire internal architecture of the OLE2 format to find every byte of unused space, regardless of whether a known steganography tool was used.
- **Modern Era (Automated Statistical Defense)**: The latest evolution involves using high-level statistics (Kurtosis) to automate detection. This marks a shift from manual binary "carving" to an ecosystem where the forensic software can autonomously flag documents based on the mathematical "structure" of their unused space.

33 Overview of Digital Forensics and Anti-Forensics Techniques,"

1. Comprehensive Taxonomy

The research classifies the field into a hierarchical structure based on the **investigative phase** and the **attacker's counter-strategy**.

- **Digital Forensic Process (Methodology)**:
  - **Acquisition/Collection**: Legally seizing and protecting digital data to ensure its original state is preserved.
  - **Examination/Analysis**: Using specialized tools to extract and interpret data, including deleted or hidden files.
  - **Reporting/Presentation**: Documenting findings for legal or administrative testimony.
- **Anti-Forensics (Application Domain)**:
  - **Data Hiding**: Making data invisible to standard tools (e.g., steganography, encryption, or slack space usage).
  - **Artifact Wiping**: Permanent deletion of data using specialized wiping software to ensure it cannot be "carved" or recovered.
  - **Trail Obfuscation**: Altering system logs, timestamps, and headers to mislead investigators.
  - **Attacking the Tool/Process**: Exploiting vulnerabilities in forensic software (like EnCase or FTK) to crash the system or provide false results.

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2. Comparative Analysis

The study provides a technical comparison of anti-forensic methods and the corresponding defensive protocols employed by investigators.

AF Technique	Goal	Forensic Countermeasure	Security Impact
Encryption	Data confidentiality	Brute force or "Live" memory forensics.	High; often a complete "dead-end" without keys.
Steganography	Data invisibility	Statistical steganalysis.	Medium; difficult to detect without a "clean" original.
Wiping	Irrecoverability	File system journal and log analysis.	High; prevents physical bit-stream recovery.

Program Packing	Obfuscating code	Sandbox execution and static/dynamic analysis.	Medium; slows down malware investigation.

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3. Critical Gap Identification

The research identifies several fundamental "bottlenecks" and missing elements in the current digital forensic landscape:

- **The "Distributed Logging" Problem**: As forensic targets move to the Internet of Things (IoT), there is a critical gap in standardized, distributed logging schemes that can survive local device tampering.
- **Automated AF Tools**: A major gap exists between the automation of anti-forensics (which is often script-based and fast) and the forensic response (which often remains manual and slow).
- **The Volatile Data Crisis**: Current literature still struggles with a "standardized" way to capture live memory data without altering the state of the machine—a gap frequently exploited by "Analysis Prevention" techniques.
- **Privacy vs. Evidence**: There is an unresolved legal and technical gap regarding the balance between user privacy rights and the intrusive nature of digital forensic acquisition.

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4. Evolution of Solutions

The narrative of digital forensics has evolved from simple data recovery to a complex, multi-layered "war" against anti-forensic automation.

- **Initial Stage (Manual Discovery)**: The field began with basic bit-by-bit copying of hard drives and manual verification of file signatures.
- **Intermediate Stage (Tool Standardization)**: The development of professional suites (EnCase, Sleuth Kit) allowed for rapid searching and automated file carving.
- **The Anti-Forensic Response**: Attackers responded by creating specialized tools to target these suites, leading to an era where forensics had to become "anti-anti-forensics" (e.g., verifying tool integrity before use).
- **Modern Era (Intelligent & Distributed Forensics)**: The current evolution involves moving toward **proactive forensics** and **distributed logging for IoT**. Instead of just looking at one computer, the field is evolving into a complex ecosystem where investigators must correlate traces across cloud servers, mobile devices, and memory artifacts simultaneously.

34 Overview on Hiding Data in Microsoft Office Documents

1. Comprehensive Taxonomy

The research classifies data hiding techniques based on the **layer of the OOXML structure** where the information is embedded.

- **ZIP-Level Methods**: Utilizing the container properties of the DOCX/XLSX/PPTX archive.
    - **ZIP Comments**: Adding text to the comment field of the ZIP header.
    - **Restricted Fields**: Using "Extra Fields" within the ZIP structure that are ignored by standard Office readers.
- **XML-Level Methods**: Modifying the markup language that defines the document content.
    - **Additional XML Fields**: Creating custom XML files or tags within the archive that do not affect the visible document.
- **Application-Level Methods**: Using built-in Microsoft Office features to obscure data.
    - **Hidden Text**: Utilizing the "Hidden" font attribute.
    - **Tracking Text**: Using the "Track Changes" feature to hide data within deleted or modified text fragments.
    - **File Embedding**: Hiding a secondary file as an OLE object within the primary document.

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2. Comparative Analysis

The study evaluates these six embedding algorithms based on their **stability** (resistance to being lost when the document is edited) and their **detectability**.

Algorithm	Type	Stability	Main Advantage / Disadvantage
ZIP Comments	ZIP	Low	High capacity; easily lost if resaved in Office.
Restricted Fields	ZIP	Medium	Difficult to detect; requires specialized ZIP parsers.
File Embedding	App	High	Standard feature; significantly increases file size.
Additional XML	XML	High	Very stable; remains even after standard editing.
Hidden Text	App	High	Simple to implement; easily found by "Show/Hide ¶".

Tracking Text	App	Medium	Good for small strings; lost if "Accept All Changes" is used.

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3. Critical Gap Identification

The research identifies several technical "blind spots" in current document security and forensic literature:

- **The "Structural Blindness" Gap**: Most forensic and anti-virus tools focus on the **visual content** or **macro-malware**, leaving a critical gap in the detection of data hidden in the ZIP-level "Extra Fields" or custom XML tags.
- **The Stability-Capacity Trade-off**: There is a missing standardized protocol for hiding *large* amounts of data that can survive a standard "Save As" operation in Microsoft Office, as many ZIP-level metadata fields are wiped during resaving.
- **Automated Leakage Detection**: The paper highlights a gap in the development of criteria for "revealing data loss"—automated systems that can alert a user if a document contains "invisible" attachments before it is sent.

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4. Evolution of Solutions

The narrative of document-based data hiding has transitioned from simple visual tricks to exploiting the complexity of modern container formats.

- **Initial Stage (Visual Obfuscation)**: Early methods relied on simple application features like "White-on-White" text or hidden font attributes. These were easily defeated by selecting all text.
- **Intermediate Stage (OLE & Binary Hiding)**: With the OLE2 format (legacy .doc), data was hidden in unused "slack space" or property streams. This required deep binary analysis to detect.
- **The OOXML Shift (2007–Present)**: The move to the ZIP-based Open XML standard opened a new thematic era of "Archive Steganography," where data is hidden in the relationship between XML files and ZIP headers.
- **Modern Era (Weaponized Documents)**: The current evolution involves "Weaponized Office Documents," where steganography is used to hide malware components (e.g., Ursnif) within legitimate-looking files to bypass email security gateways. The field is now moving toward **structural archive analysis** as a standard forensic requirement.

35 Review of the algorithms steganography in PDF documents and analysis,

1. Comprehensive Taxonomy

The research classifies PDF steganography based on the **structural level** of the document where data is hidden.

- **Structure-Based Methods**: Utilizing the internal organization of the PDF file format.

    - **Incremental Updates**: Hiding data by appending new objects to the end of the file, which standard viewers may ignore.
    - **Unused Objects**: Placing data in objects that are defined in the file but never referenced by the document's pages.
- **Operator-Based Methods**: Modifying the specific instructions (operators) used to render the document content.
    - **Text Positioning (Tj/TD Operators)**: Altering the spacing between characters or words.
    - **Color Space Manipulation**: Hiding data within the color definitions of text or graphics.
- **Metadata-Based Methods**: Embedding information in the document's descriptive properties.
    - **Document Information Dictionary**: Using fields like "Author" or "Keywords" to store hidden strings.
    - **XMP Metadata**: Utilizing the Extensible Metadata Platform streams for data storage.

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2. Comparative Analysis

The study analyzes various PDF steganography algorithms based on their **statistical impact** and **detection resistance**.

Algorithm Type	Capacity	Detectability	Primary Vulnerability
Metadata Hiding	Low	High	Easily flagged by simple metadata parsers.
Incremental Updates	High	Medium	File size increases significantly; detectable by structural analysis.
Tj Operator Spacing	Moderate	Low	Requires complex statistical analysis of character positioning to detect.
Hidden Layers	Moderate	Medium	Can be revealed by certain PDF editing tools.

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3. Critical Gap Identification

The research identifies several unresolved challenges and "blind spots" in the current literature and forensic practice:

- **The "Structural Integrity" Gap**: There is a lack of standardized detection algorithms that can identify hidden data without a "clean" original document for comparison, particularly when structural changes are minimal.
- **Statistical Baseline Deficit**: A major gap exists in defining the "normal" statistical characteristics of PDF operators (like Tj). Without a baseline for how different PDF creators position text, it is difficult to identify steganographic anomalies.
- **Cross-Platform Consistency**: The paper highlights a gap in how steganography survives across different PDF versions (e.g., 1.4 vs. 2.0) and different rendering engines, which often handle "malformed" or "unused" objects differently.

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4. Evolution of Solutions

The narrative of PDF steganography has evolved from simple text-based masking to exploiting the complex object-oriented nature of the format.

- **Initial Stage (Simple Text Hiding)**: Early methods relied on making text invisible (white-on-white) or placing it behind images. These were easily defeated by "selecting all" text.
- **Intermediate Stage (Metadata and Slack Space)**: As the PDF format became an ISO standard, focus shifted to hiding data in "slack space" between objects or within the metadata dictionary.
- **Operator and Stream Sophistication**: The field evolved to manipulate the actual rendering instructions. By subtly changing the "Tj" operator values for character positioning, data could be hidden in a way that is visually identical to the original.
- **Modern Era (Automated Detection Systems)**: The current evolution (proposed in this article) involves moving toward **automated statistical detection**. By analyzing the distribution of specific PDF operators across large datasets, the field is moving toward an "intelligent defense" ecosystem that can flag anomalous files without human intervention.

36 Securing Web User Privacy with Steganalysis of Images using Deep Learning,

1. Comprehensive Taxonomy

The research classifies image steganalysis based on the **analysis framework** and the **feature extraction architecture** used to detect hidden data.

- **By Analysis Framework**:
  - **Traditional Steganalysis**: Relying on hand-crafted statistical features and ensemble classifiers to detect pixel correlations.
  - **Deep Learning Steganalysis**: Using automated neural networks to learn high-level representations of steganographic noise.
- **By Neural Network Architecture**:
  - **Convolutional Neural Networks (CNN)**: Utilizing multiple layers (convolutional, pooling, and fully connected) to extract structural anomalies.

    - **Deep Residual Networks**: Improving detection by focusing on the "residual" noise—the difference between the stego-image and a predicted "clean" version.
- **By Target Domain**:
    - **Spatial Domain Steganalysis**: Specifically targeting Least Significant Bit (LSB) embedding in raw pixel data.
    - **JPEG Domain Steganalysis**: Focusing on artifacts hidden within Discrete Cosine Transform (DCT) coefficients.

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2. Comparative Analysis

The study evaluates the proposed deep learning model against traditional statistical methods across key performance metrics.

Metric	Traditional (Rich Models)	Proposed CNN Model
Feature Extraction	Manual/Hand-crafted	Automated (Convolutional)
Scalability	Low; requires specific domain knowledge	High; adapts to various image sizes and formats
Computational Efficiency	High for small datasets	Low; requires GPU training for high performance
Detection Accuracy	Variable; struggles with low-payload embedding	**94.8%** (Mean accuracy across tested sets)

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3. Critical Gap Identification

The research identifies several fundamental "bottlenecks" and unresolved challenges in the current literature of digital image privacy:

- **The "Arbitrary Size" Gap**: Most existing CNN-based steganalysis models are designed for fixed image dimensions (e.g., 256x256). There is a critical gap in models that can effectively analyze images of arbitrary size without resizing, which often destroys steganographic traces.
- **Payload Sensitivity**: A major gap exists in detecting "low-payload" steganography (where very few bits are changed). Most current tools fail to distinguish this minor noise from natural camera sensor noise.

- **Real-time Web Monitoring**: The paper highlights a lack of standardized, high-speed steganalysis ecosystems capable of monitoring the massive volume of images uploaded to social media platforms in real-time.
- **Adversarial Robustness**: There is a missing link in research regarding how steganalysis models perform against "adversarial steganography," where the embedding process is specifically designed to fool neural networks.

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4. Evolution of Solutions

The narrative of steganalysis has transitioned from manual mathematical verification to fully autonomous, feature-aware neural ecosystems.

- **Initial Stage (Visual & Statistical Checks)**: Early solutions focused on visual artifacts and first-order statistics (e.g., histogram analysis) to find bit-plane irregularities.
- **Intermediate Stage (Rich Models & Ensemble Classifiers)**: Researchers developed "Rich Models" that used high-dimensional hand-crafted features. These required massive computational power to "ensemble" different classifiers for a final decision.
- **The Deep Learning Shift**: The introduction of CNNs allowed for the automatic learning of "noise filters." This moved the field away from manual engineering toward an era where the software "discovers" the steganographic signature itself.
- **Modern Era (Privacy-Centric Deep Steganalysis)**: The current evolution (as proposed in this study) involves **end-to-end deep learning architectures**. By integrating noise-aware layers directly into the web user privacy framework, the field is moving toward a "Smart Web Defense" ecosystem that can autonomously flag potentially harmful hidden data in online media.

37 Steganalysis of AI Models LSB Attacks,

1. Comprehensive Taxonomy

The research classifies the security of AI model sharing into a hierarchy based on the **attack vector** and the **detection methodology**.

- **By Attack Vector (Model Zoo Exploitation)**:
  - **LSB Model Steganography**: Injecting malicious payloads into the Least Significant Bits (LSB) of a pre-trained AI model's weights.
  - **EvilModel Attacks**: A specific class of attack where malware is hidden inside neural network models to bypass traditional security scanners.
- **By Steganalysis Methodology**:
  - **Supervised Learning Detection**: Training a secondary AI model to recognize the specific "statistical fingerprint" left by a steganographic injection.
  - **Unsupervised Learning Detection**: Using anomaly detection to identify weights that deviate from the expected distribution of a "clean" model without prior training on attack samples.
- **By Data Precision Format**:

- **Floating-Point Analysis**: Targeting the 32-bit or 64-bit representations of model weights where LSB manipulation occurs.

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2. Comparative Analysis

The study evaluates three distinct steganalysis methods proposed to secure the AI open-source community.

Feature	**Method 1: Supervised CNN**	**Method 2: Statistical Anomaly**	**Method 3: Unsupervised Clustering**
Detection Type	Supervised	Heuristic/Statistical	Unsupervised
Complexity	High (requires training)	Low	Moderate
Accuracy (MSB-LSB)	Near 100%	High	Moderate
Accuracy (Deep-LSB)	Low	Very Low	Low
Primary Use Case	Known attack patterns	Rapid screening	Identifying "Zero-day" stego

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3. Critical Gap Identification

The research identifies significant "blind spots" in the protection of the AI open-source ecosystem:

- **The "Deep LSB" Detection Gap**: A critical gap exists in detecting attacks that exploit the very last bits of a weight. If an attacker limits their payload to the least significant bits, current detection ability remains low.
- **Model Integrity vs. Performance**: There is an unresolved challenge in creating steganalysis that can detect hidden data without access to the "original" clean version of the model for comparison.

- **Trust in "Model Zoos"**: The paper highlights a lack of standardized security protocols for repositories like TensorFlow Hub, leaving them vulnerable to hosting "EvilModels" that appear to function normally but contain dormant malware.

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4. Evolution of Solutions

The narrative of AI security has shifted from protecting the *output* of models to protecting the *weights* themselves.

- **Initial Stage (Classic Malware)**: Early cyber risks focused on malicious code files. AI models were seen as "black box" data files that were inherently safe.
- **Intermediate Stage (EvilModel 1.0/2.0)**: Attackers discovered that the massive number of parameters in a neural network (millions of weights) provided perfect "slack space" to hide full malware binaries.
- **The Steganalysis Response**: The field moved from simple file hashing to **weight-level inspection**. Researchers began applying image steganography principles to the numerical values within neural networks.
- **Modern Era (Zero-Trust AI Models)**: The current evolution (proposed in this work) involves **AI-on-AI steganalysis**. By using both supervised and unsupervised methods, the field is moving toward a "Zero-Trust" ecosystem where models are validated at the bit-level before being integrated into production systems.

38 Steganographic Algorithm Injection in Image Information Systems used in Healthcare Organizations

1. Comprehensive Taxonomy

The research classifies the protection of healthcare image data based on the **injection methodology** and the **authentication scheme** used to verify clinical integrity.

- **By Methodology (Information System Integration)**:
  - **External Security**: Using traditional database security and firewalls to protect images within an Image Information System (IIS).
  - **Injected Security (Proposed)**: Injecting authentication metadata (author ID and image properties) directly into the image pixels to ensure protection even when the image is exposed on the web.
- **By Authentication Domain (Healthcare Specific)**:
  - **Patient/Author ID**: A unique, encrypted identifier generated for healthcare professionals and patients.
  - **Clinical Metadata**: Information regarding the medical image itself, such as modality or capture date, used for forensic verification.
- **By Cryptographic Method**:
  - **Public-Key Functions**: Utilizing intractable mathematical problems (like factorization) to secure the generated IDs before embedding.
  - **Digital Signatures**: Using forward-secure digital signature schemes to ensure the metadata cannot be tampered with.

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2. Comparative Analysis

The study evaluates the proposed "Algorithm Injection" against traditional web-based image exposure risks.

Feature	Traditional IIS Security	Proposed Injected Steganography
Protection Scope	Internal (Database/Firewall)	Permanent (Attached to Pixels)
Web Exposure Risk	High; images lose "author" context after copy	Low; authenticity is embedded in the data
Authentication Type	External Logins	Cryptographic IDs and Digital Signatures
Primary Domain	General Information Systems	Healthcare Organizations (IIS)

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3. Critical Gap Identification

The research identifies specific technical and systemic gaps in the current management of medical imagery:

- **The "Web Copy" Identity Gap**: A critical gap exists where medical images, once exported from a secure internal system to the web for consultation, lose their connection to their original author and authenticity.
- **Metadata Disassociation**: Traditional systems often store "metadata" separately from the "image." This creates a gap where the image can be tampered with or misidentified once separated from its database record.
- **The Intruder Interest Problem**: There is an unresolved challenge regarding "outer intruders" who specifically target healthcare IIS to possess and misuse sensitive patient data for their own interest.

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4. Evolution of Solutions

The narrative of healthcare data security has moved from protecting the "container" to protecting the "content" itself.

- **Initial Stage (Boundary Defense)**: Early solutions focused on securing the database and the network perimeter. Images were considered safe as long as they remained inside the hospital network.

- **Intermediate Stage (Standard Steganography)**: Researchers began using basic data hiding (like LSB) to put small watermarks on images, but these often lacked strong cryptographic protection.
- **Cryptographic Integration**: The field evolved to combine steganography with **Public-Key Cryptography** and **Digital Signatures** (e.g., Rabin's functions), ensuring that hidden data was not only invisible but also mathematically secure.
- **Modern Era (Intelligent IIS Injection)**: The current evolution involves **Steganographic Algorithm Injection**. Instead of a simple watermark, the system now injects a complex "Identity Scheme" directly into the clinical data. This marks the shift toward a "Self-Authenticating" healthcare ecosystem where every image carries its own encrypted forensic record, regardless of where it is stored or shared.

39 The Value of Metadata in Digital Forensics

1. Comprehensive Taxonomy

The research classifies metadata based on its **visibility** to the average user and its **functional role** in establishing evidence.

- **By Visibility**:
  - **Embedded (Hidden) Metadata**: Information not visible when viewing data in common forms (e.g., a Word document or an image).
  - **Visible Metadata**: Properties that can be discovered easily through standard file viewing, though often overlooked by non-experts.
- **By Investigative Domain**:
  - **File/Document Metadata**: Used to discover the specific properties, authorship, and modification history of a file.
  - **Network Activity Metadata**: Used to discover the origins, destinations, and timing of digital communications.
- **By Legal Utility**:
  - **Corroborative Evidence**: Metadata that provides proof of collaboration between groups of people who may be unaware that such "tracking" information is stored within their documents.
  - **Validation Data**: Information used to validate a legal argument or verify the authenticity of relevant evidence.

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2. Comparative Analysis

The study highlights the technical and legal differences between analyzing the "content" of digital evidence versus its "metadata".

Feature	Content Analysis	Metadata Analysis

Primary Focus	What the document says or what the image shows.	How, when, and by whom the file was handled.
Accessibility	Readily apparent to any viewer.	Often hidden; requires forensic tools for access.
User Awareness	High; users generally know what they wrote.	Low; users are often unaware of stored properties.
Investigative Value	Provides direct evidence.	Provides collaborative and forensic validation.
Legal Risk	Content can be easily falsified.	Metadata is harder to alter without leaving traces.

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3. Critical Gap Identification

The research identifies several technical and educational gaps that can lead to "misdirection" in digital investigations:

- **The "Legal Practitioner" Knowledge Gap**: A significant gap exists where legal professionals often do not understand how metadata is collected or its relationship to corroborating evidence, which can lead to the failure to identify pertinent proof.
- **Tool Inconsistency**: The use of unsuitable tools or application errors during collection can result in the misrepresentation or even falsification of evidence.
- **The Exculpatory Evidence Blind Spot**: Failure to report metadata that could clear a suspect (exculpatory evidence) is an unresolved challenge in the standard investigative process.
- **Systemic Errors**: Errors during the evidence collection process can lead to a lack of validation, rendering the data inadmissible in court.

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4. Evolution of Solutions

The narrative of digital forensics has evolved from simple file recovery to the sophisticated validation of "hidden" digital properties.

- **Initial Stage (Content-Only Focus)**: Early forensics focused on what was visible on the screen. Investigators and legal teams rarely looked "behind" the file for hidden properties.

- **Intermediate Stage (Metadata Discovery)**: As document formats became more complex (e.g., MS Word, PDFs), investigators began using specialized tools to reveal authorship and timestamps that were "not visible when viewing data" in its standard form.
- **The Validation Era**: The field shifted toward using metadata specifically to **validate legal arguments**. Metadata became a tool not just for discovery, but for ensuring the integrity of the entire forensic process.
- **Modern Era (Integrated Legal-Forensic Synergy)**: The current evolution involves a "collaborative" approach where the digital investigator and the legal practitioner must work in tandem. The field is moving toward a standard where the **validation of metadata** for court presentation is considered as important as the evidence itself, ensuring a robust chain of custody and protection against misdirection.

40 Validity of Forensic Evidence using Hash Function,

1. Comprehensive Taxonomy

The research classifies the use of hash functions based on the **investigative phase** and the **security properties** required for forensic validity.

- **By Investigative Phase**:
  - **Evidence Gathering**: Identifying and seizing digital media such as surveillance systems, hard disks, and pen drives.
  - **Forensics Copying**: Creating an exact bit-stream duplicate of original evidence to prevent accidental destruction or alteration.
  - **Verification**: Using hash values at any point in the timeline to ensure the evidence currently being analyzed is identical to the original forensic copy.
- **By Security Requirement**:
  - **Integrity Assurance**: Proving that the evidence has not been subject to unauthorized manipulation.
  - **Collision Resistance**: The mathematical requirement that it should be impossible to find two different inputs that produce the same hash output.

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2. Comparative Analysis

The study evaluates the technical strengths and vulnerabilities of widely used forensic hash algorithms.

Feature	**MD5**	**SHA-1**	**SHA-2 / SHA-3**
Bit Length	128-bit	160-bit	256-bit to 512-bit

Current Status	Considered broken/insecure	Increasingly considered weak	Current gold standard for security
Vulnerabilities	Collision, Birthday, and Rainbow attacks	Susceptible to collision finding	Theoretically resistant but slower on some CPUs
Forensic Utility	Used for rapid legacy checks	Historically the most widely used	Recommended for modern legal validity

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3. Critical Gap Identification

The research identifies several technical and procedural "blind spots" in the application of hashing to digital forensics:

- **The "Collision Resistance" Debate**: A significant gap exists between cryptographic experts who insist MD5 is unusable and practitioners who argue that real-time collisions are too rare to impact daily forensic work.
- **Performance vs. Security**: A major gap in SHA-3 adoption is its perceived "slowness" on general-purpose processors compared to its predecessors, leading many investigators to stick with older, less secure algorithms.
- **The Migration Lag**: There is an unresolved challenge in the forensic community regarding the slow migration to newer, safer algorithms even after vulnerabilities in MD5 and SHA-1 have been proven.

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4. Evolution of Solutions

The narrative of digital evidence validation has shifted from physical protection to mathematical verification.

- **Initial Stage (Original Handling)**: Early forensics often involved working directly on original evidence, which led to high risks of data destruction or alteration.
- **Intermediate Stage (Forensic Imaging & MD5)**: The field moved toward creating "forensic copies" and using MD5 to provide a "fingerprint." This allowed investigators to prove that the copy used in the lab matched the original seized at the scene.
- **The Security Crisis**: As mathematical attacks (like the Birthday attack) became more feasible against MD5, the field evolved to adopt the SHA family (Secure Hash Algorithm) for higher reliability.
- **Modern Era (High-Bit Integrity & SHA-3)**: The current evolution involves moving toward SHA-2 and SHA-3. The field is shifting toward a standard where the **validity of forensic evidence** is tied to algorithms with high collision resistance, ensuring that even sophisticated unauthorized manipulations can be mathematically detected.

Tab 2

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